

★ WEEK 4 · PART 1 of 3

ARIMA: Integrating the Trend

Extending ARMA to Non-Stationary Data

Differencing · the order of integration d · ARIMA(p,d,q) · forecasting trends

ARMA needs a stationary series — but most real data trends. ARIMA closes that gap with one idea: difference the series until it is stationary, model the result, then integrate the forecast back.

1

PART 1 • ARIMA

From ARMA to ARIMA

The 'I' for integration: how differencing removes a stochastic trend, what the order d means, and how to forecast a trending series honestly.

The Airline That Taught Us to Forecast — Part 1

The dataset that showcased ARIMA

Monthly international airline passengers, 1949–1960, rose relentlessly AND swung in a yearly rhythm that grew with the level — a series no stationary model could touch.

Two problems at once

- A strong upward trend: the mean never sits still.
- A repeating yearly season whose swings grow over time.
- Plain ARMA assumes neither — it needs a stationary series.

Why it became famous

- Box and Jenkins used it as the showcase for their method.
- It motivated both the 'I' (this part) and the seasonal 'S' (Part 2).
- The 'airline model' is still a textbook benchmark today.

The Airline That Taught Us to Forecast — Part 2

The ARIMA idea, and the lesson

The fix was disarmingly simple: difference the series to strip out the trend, model what remains with ARMA, then add the trend back when forecasting.

What ARIMA added

- One integer, d : how many times to difference.
- Everything else is the Week-3 ARMA machinery, reused.
- Trend handled by differencing; season handled next week.

The lesson for us

- Do not fight a trend — difference it away, then model.
- The order d is a modeling choice you must justify.
- This one idea unlocks the vast world of trending data.

The Gap ARMA Leaves

Three reasons ARMA alone is not enough

1

ARMA assumes stationarity

Its whole theory — the ACF, the estimates, the forecasts — rests on a constant mean and variance. Real economic series rarely oblige.

2

Most real series trend

Prices, GDP, sales, and passenger counts drift upward over time. Applying ARMA directly courts the spurious problems of Week 2.

3

Differencing bridges the gap

We already know (Week 2) that differencing removes a stochastic trend. ARIMA simply bolts that step onto ARMA and formalizes it as the order d .

ARIMA in one line: difference to stationarity, model with ARMA, integrate the forecast back.

The 'I' in ARIMA: Integration

Where the third letter comes from

A series is 'integrated' if it must be differenced to become stationary. ARIMA layers exactly that step — the Integration order d — on top of an ordinary ARMA.

The name, decoded

- AR: autoregression on past values (Week 3).
- MA: moving average of past shocks (Week 3).
- I: the number of differences, d , to reach stationarity.

Why 'integrated'

- Differencing is like a derivative; undoing it is integration.
- An $I(1)$ series is the running sum of a stationary one.
- So we difference to model, then integrate to forecast.

Differencing, Precisely

The single operation at the heart of ARIMA

$$\Delta X_t = X_t - X_{t-1} = (1 - L) X_t \quad (\text{first difference})$$

What it does

- Replaces each value with its change from the period before.
- In lag-operator form it is simply $(1 - L)$ applied to X_t .
- One pass removes a linear stochastic trend.

Why it works

- A random walk's differences are white noise — instantly stationary.
- Differencing a drifting series leaves a stationary series plus a constant.
- That constant becomes the drift in the forecast.

The Order of Integration, d

How many differences does the series need?

$$X_t \sim I(d) \iff (1 - L)^d X_t \text{ is stationary (ARMA)}$$

The ladder, revisited

- $d = 0$: already stationary — plain ARMA (Week 3).
- $d = 1$: one difference (most trending series, prices).
- $d = 2$: two differences (rare — some accelerating series).

Choosing d well

- d is almost always 0, 1, or 2 — seldom more.
- Under-differencing leaves a trend; over-differencing adds noise.
- Tests and the ACF (a few slides on) pin it down.

ARIMA(p, d, q)

Three integers describe the whole model

$$\underbrace{\phi(L)}_{\text{AR } p} \underbrace{(1 - L)^d}_{\text{I } d} X_t = c + \underbrace{\theta(L)}_{\text{MA } q} \varepsilon_t \quad \text{ARIMA}(p, d, q)$$

Reading the three numbers

- p: the AR order — memory of past levels.
- d: the number of differences to reach stationarity.
- q: the MA order — memory of past shocks.

How it is fit

- Difference the series d times to make it stationary.
- Identify and estimate an ARMA(p, q) on the result.
- The software does all three steps in one call.

The Random Walk Is ARIMA(0,1,0)

Old friends, in the new language

$$(1 - L) X_t = \varepsilon_t \iff \text{random walk} = \text{ARIMA}(0, 1, 0)$$

The special cases

- ARIMA(0,1,0): a random walk — difference once, model nothing.
- ARIMA(0,1,0) with a constant: a random walk WITH drift.
- ARIMA(p,0,q): an ordinary stationary ARMA.

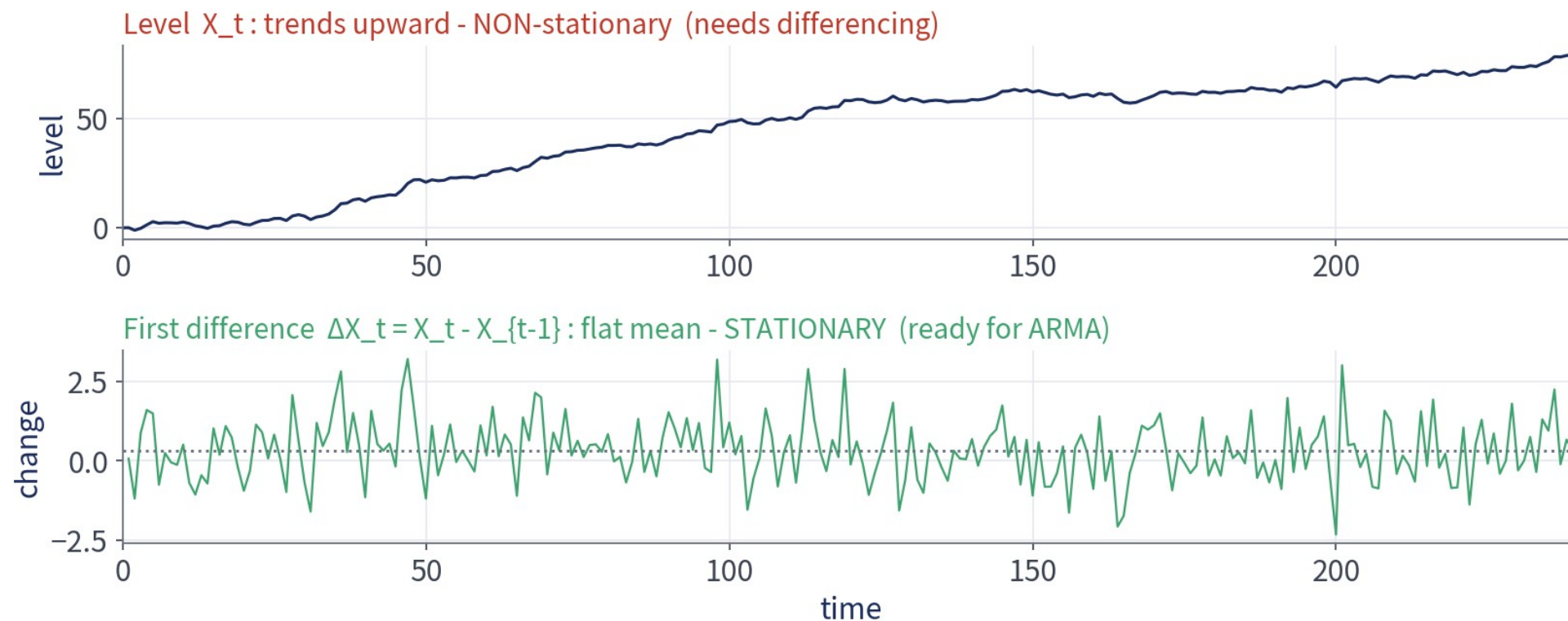
Why this is satisfying

- The unit-root series of Week 2 now has a name and a slot.
- ARIMA unifies random walks, trends, and ARMA in one family.
- Every model this course has met is an ARIMA special case.

One Difference Turns Trend Into Stationarity

Top: a trending, non-stationary level. Bottom: its first difference fluctuates around a flat mean — exactly the stationary input ARMA requires.

The 'I' in ARIMA: one difference turns a trending series into a stationary one



When Should You Difference?

Reading the series for a stochastic trend

Difference when the series has a stochastic trend — a wandering level with no fixed mean and a slowly-decaying ACF. That is the unit-root signature from Week 2.

Signs you need $d \geq 1$

- The level wanders with no mean it returns to.
- The ACF decays very slowly, staying near 1.
- ADF fails to reject a unit root; KPSS rejects stationarity.

Signs you do NOT

- The series already fluctuates around a stable mean.
- Its ACF dies out quickly.
- Differencing here would over-difference (adds noise).

How Many Differences? Let the Tests Decide

Reusing the Week-2 toolkit

The procedure

- Run ADF and KPSS on the level; if non-stationary, difference once.
- Re-test the differenced series; repeat only if still non-stationary.
- Stop as soon as the tests agree it is stationary — that is d .

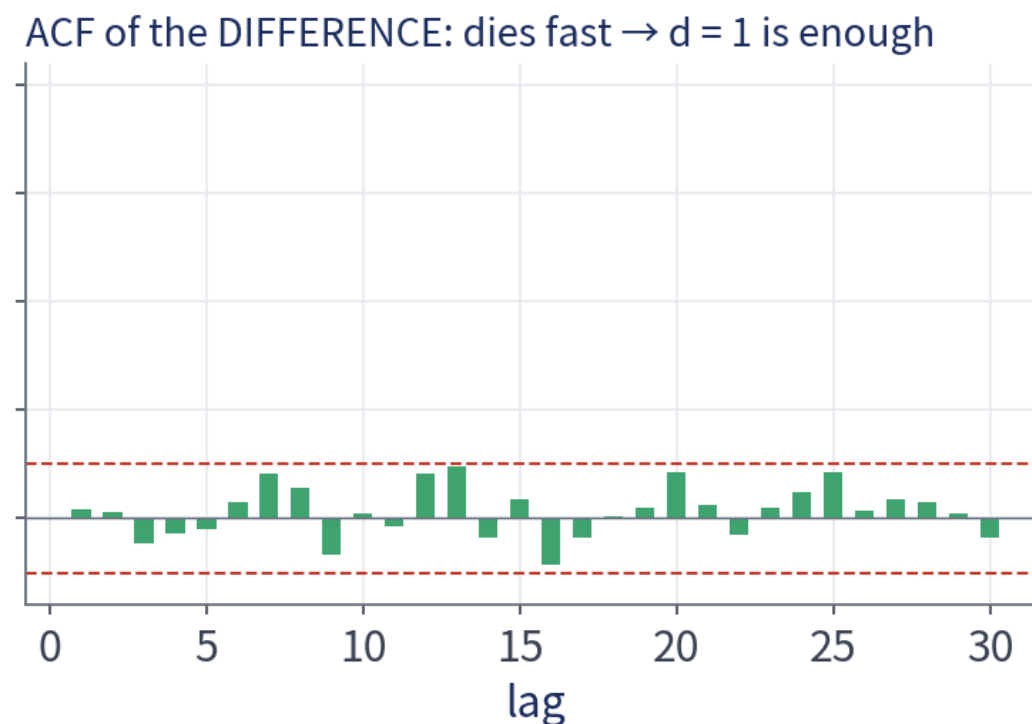
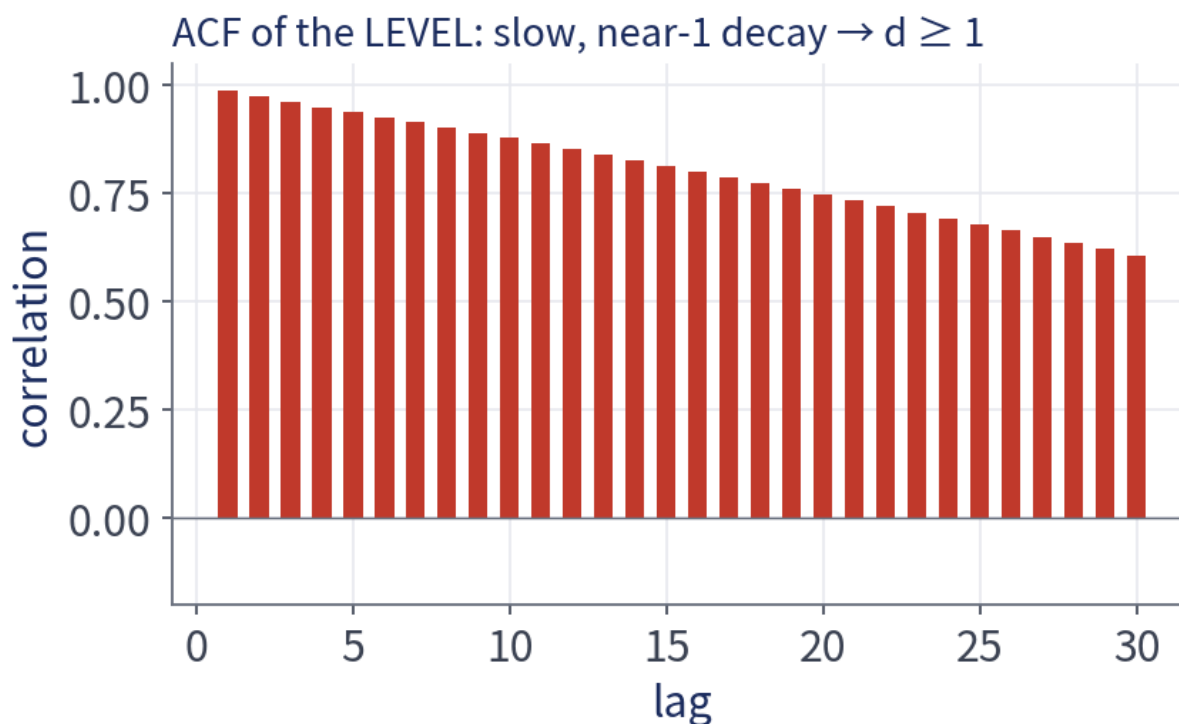
Practical guidance

- Almost every financial or macro series needs $d = 0$ or $d = 1$.
- $d = 2$ is uncommon; $d \geq 3$ almost never justified.
- When unsure, prefer the smaller d and check the residuals.

Choosing d From the ACF

A slowly-decaying ACF on the level (left) says 'difference'. After one difference the ACF dies out fast (right) — $d = 1$ is enough; a further difference would over-do it.

Choosing d: difference until the ACF stops decaying slowly



Over-Differencing, Again

Too much of a good thing

Differencing a series that is already stationary injects a large negative lag-1 autocorrelation and inflates the variance — the tell-tale sign you have differenced once too many.

The symptoms

- A strong negative spike at lag 1 in the differenced ACF.
- Noticeably larger variance than the original series.
- An MA term that keeps trying to undo the extra difference.

The remedy

- Choose the smallest d that achieves stationarity.
- Confirm with ADF/KPSS rather than differencing on reflex.
- If in doubt, difference less, not more.

Difference vs Detrend — Still Matters

ARIMA is for stochastic trends, not deterministic ones

Difference-stationary (use ARIMA)

- A stochastic trend — a unit root, random-walk-like.
- Shocks are permanent; the level wanders.
- Cure: difference (this is the 'I' in ARIMA).

Trend-stationary (detrend instead)

- A fixed deterministic trend, $a + b \cdot t$.
- Shocks are temporary; the series hugs the line.
- Cure: subtract the trend, then model the residual with ARMA.

Modeling the Differenced Series

Once stationary, it is just ARMA

After differencing d times, the hard part is over: identify p and q on the differenced series using the Week-3 ACF/PACF cheat sheet, exactly as before.

What carries over

- The ACF/PACF fingerprints identify p and q as usual.
- AIC/BIC still select among candidates.
- Residual diagnostics still gate the final model.

The only new step

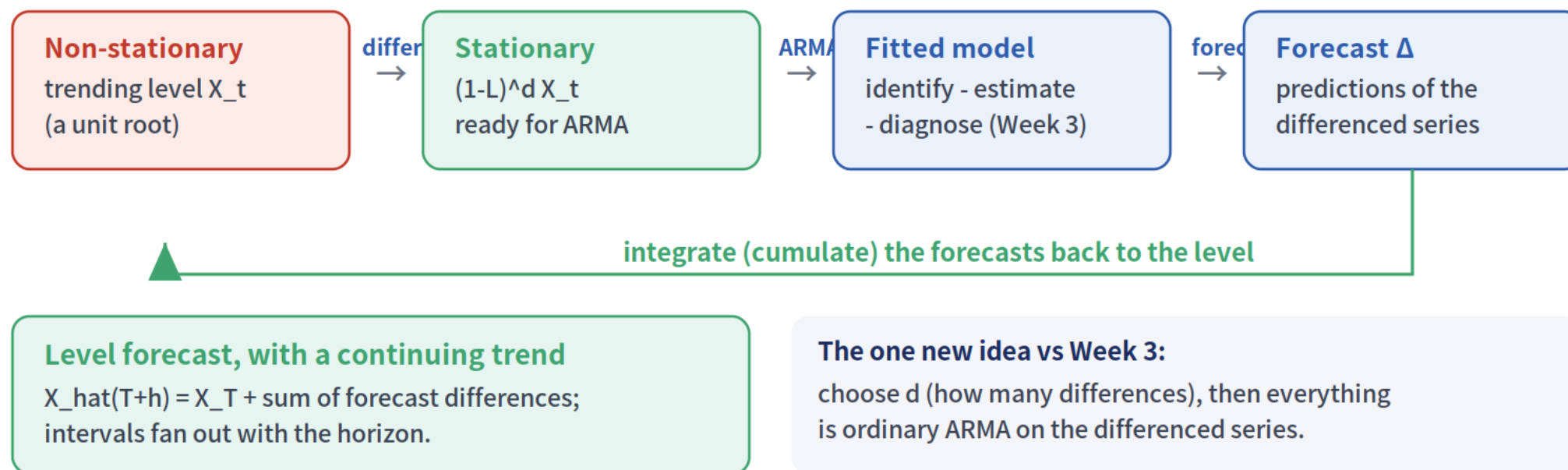
- You chose d first, and you will integrate the forecast back.
- Everything between is ordinary ARMA on the difference.
- So Week 3's entire toolbox applies unchanged.

The ARIMA Pipeline

Difference until stationary, fit an ARMA, forecast the differences, then integrate those forecasts back up to the level — trend and all.

The ARIMA Pipeline

ARIMA = difference until stationary, fit an ARMA, then integrate the forecast back to levels.



Building an ARIMA, Step by Step

Five steps from raw level to forecast

1 · Test

pick d

ADF/KPSS on
the level;
difference until
stationary.

2 · Identify

p and q

Read the
ACF/PACF of the
differenced
series.

3 · Estimate

fit

Fit
ARIMA(p,d,q);
compare by
AIC/BIC.

4 · Diagnose

residuals

Confirm the
residuals are
white noise.

5 · Forecast

integrate

Predict, then
cumulate back
to the level.

Only steps 1 and 5 are new; steps 2–4 are the Week-3 loop, unchanged.

Forecasting: Integrate the Differences Back

The trend continues, not reverts

$$\hat{X}_{T+h} = X_T + \sum_{i=1}^h \widehat{\Delta X}_{T+i} \quad (\text{integrate the forecasts back})$$

How it works

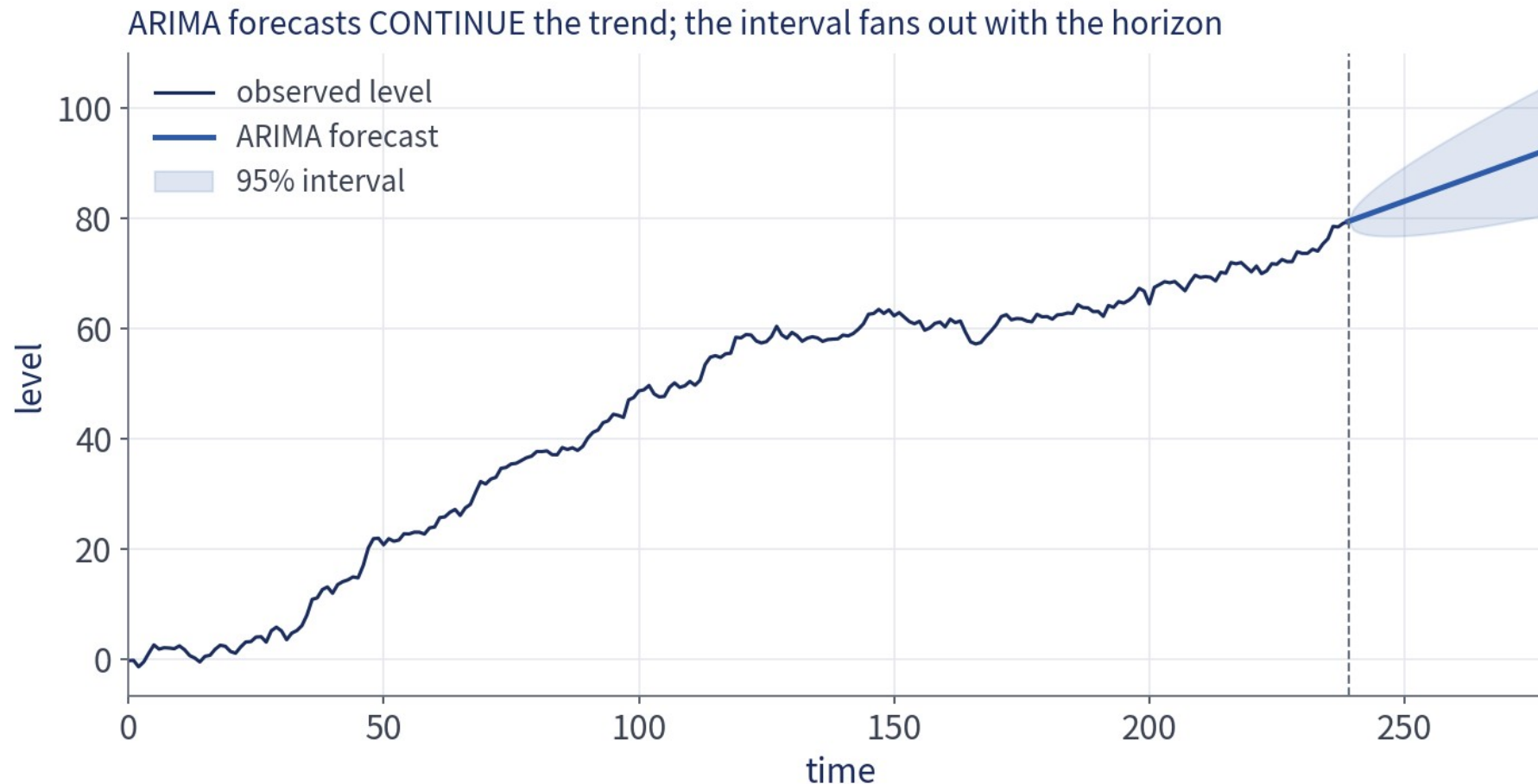
- The model forecasts the DIFFERENCES (the stationary part).
- Cumulate them onto the last observed level to get level forecasts.
- Any drift term makes the forecast keep trending.

How it differs from ARMA

- A stationary ARMA forecast reverts to its mean (Week 3).
- An ARIMA forecast instead CONTINUES the trend.
- The intervals fan out faster, reflecting the unit root.

An ARIMA Forecast Continues the Trend

Unlike a stationary ARMA, the point forecast extends the trend from the last level, and the interval fans out with the horizon — the footprint of the unit root.



ARIMA vs a Deterministic-Trend Model

Two ways to handle a trend — with different forecasts

Approach	Trend it assumes	How to handle it	Long-horizon forecast
ARIMA (difference)	stochastic (unit root)	difference d times	continues from the last level; wide bands
Detrend + ARMA	deterministic line $a+b \cdot t$	subtract the fitted trend	reverts to the line; narrower bands

Choosing wrong biases every long-horizon forecast — the same lesson as Week 2, now for forecasting.

Difference to Stationarity, Then Fit ARIMA

Direct Claude Code through the choose-d workflow

GOAL Take a trending series, use tests to choose d , fit an ARIMA, and forecast the level with honest intervals.

[INPUT] Prompt · Claude Code

Write and run `arima_lab.py` that:

1. downloads a trending series (e.g. a stock PRICE level, or airline-passenger data);
 2. runs ADF/KPSS on the level and on the first difference to choose d ;
 3. identifies (p, q) from the differenced ACF/PACF, fits $\text{ARIMA}(p, d, q)$, and reports AIC/BIC and the residual Ljung-Box;
 4. forecasts 20 steps ahead on the LEVEL with a 95% interval, and plots it.
- Tell me the chosen (p, d, q) and why.

• Expected $d = 1$ for most price levels.

- The forecast should continue the trend, not flatten.
- Compare it to a plain ARMA on the level — see the difference.
- Keep it; Part 2 adds the seasonal terms.

Deliverable: `arima_lab.py` + a level forecast with its fanning interval.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1	A price level is $I(1)$. What are its p , d , q if the difference is white noise?	(0, 1, 0) random walk
Q2	Your differenced series shows a big negative lag-1 ACF. What went wrong?	Over-differenced
Q3	How does an ARIMA forecast of a trend differ from an ARMA forecast?	Continues, not reverts

All three follow from differencing, the order d , and integrating the forecast back.

Three Pitfalls to Avoid

Where ARIMA modeling goes wrong

Common mistakes

- "When in doubt, difference more."
- "ARIMA and a deterministic trend give the same forecast."
- "Just fit ARMA to the price level directly."

The reality

- Over-differencing injects a negative lag-1 ACF and inflates variance.
- Stochastic and deterministic trends forecast very differently.
- Fitting ARMA to a unit-root level is the spurious trap (Week 2).

Three Things to Carry Forward

Part 1 in one screen

1

ARIMA = difference, then ARMA

The 'I' is just an integer d : difference the series d times to reach stationarity, then apply the entire Week-3 ARMA toolbox to the result.

2

Choose d with tests, not reflex

Use ADF/KPSS and the ACF to pick the smallest d that yields stationarity. Under-differencing leaves a trend; over-differencing adds noise.

3

ARIMA forecasts continue the trend

Forecast the differences, then integrate back: the point forecast extends the trend and the interval fans out — the signature of a unit root.

Next — Part 2: Seasonality and SARIMA, adding the repeating yearly rhythm the airline data showed.

★ WEEK 4 · PART 1 · PRACTICE

The ARIMA Model, Worked by Hand

Difference a Price Into Stationarity, Then Forecast the Trend

The 'I' in ARIMA, the order d , and integrating back

Part 1's core operation, with the arithmetic shown. We difference a trending price series until it's stationary, read off the order of integration, then forecast the level by integrating the differences back - watching the stochastic trend continue rather than revert.

1P

PART 1 · APPLICATIONS

One Difference Makes It Stationary

Case A: difference a price series and confirm the changes are stationary while the level trends - the order of integration is one. Case B: identify the ARIMA order, forecast the differenced series, and integrate back so the trend continues. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

The 'I' is integration - differencing

ARIMA extends ARMA to trending series by differencing them first. Each difference turns a level into a change.

2

The order d counts the differences

How many differences make the series stationary. A random walk needs one, so it's ARIMA(0,1,0).

3

Forecasting integrates the differences back

Model the stationary differences, then cumulate them - so the stochastic trend continues rather than reverting.

Everything here is a calculator exercise: one difference, one order, one integrated forecast.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a trending price series.
- Difference it, or forecast and integrate back.
- Then read the trend.

What you should watch

- How differencing removes the trend.
- How the order d is decided.
- How the integrated forecast keeps trending.

Differencing to Stationarity

A trend removed in one step

A price series wanders upward - non-stationary, with a stochastic trend. ARIMA's first move is to difference it: replace each level with its change from the day before. The differences are the daily price moves, and they fluctuate around a small constant with no trend.

We compute them and read off the order of integration

The setup

- Eight daily prices trending 100 to 113.
- Clearly non-stationary in level.
- Candidate for one difference.

What we'll do by hand

- Take first differences.
- Check they're stationary.
- Confirm the order d is one.

Differencing, Precisely

Turn a level into a change

$$\Delta y_t = y_t - y_{t-1} \quad (\text{the operation at the heart of ARIMA: turn a level into a change})$$

What it does

- Each value becomes its change from before.
- A linear trend becomes a constant.
- A stochastic trend becomes stationary noise.

In finance

- Differencing a price gives the price change.
- Or, in logs, the return.
- Prices are $I(1)$; returns are $I(0)$.

Take the Differences

The level trends, the changes don't

price	100	103	101	105	108	106	110	
difference	-	+3	-2	+4	+3	-2	+4	+

The prices climb from 100 to 113, but their differences - +3, -2, +4, and so on - hover around a small positive drift of about 1.9 with no trend of their own. One difference has made the series stationary.

The Order of Integration

And difference versus detrend

prices 100, 103, 101, 105, ... $\Rightarrow \Delta y = 3, -2, 4, 3, -2, 4, 3$ (drift 1.86, no trend $\Rightarrow d=1$)

Reading d

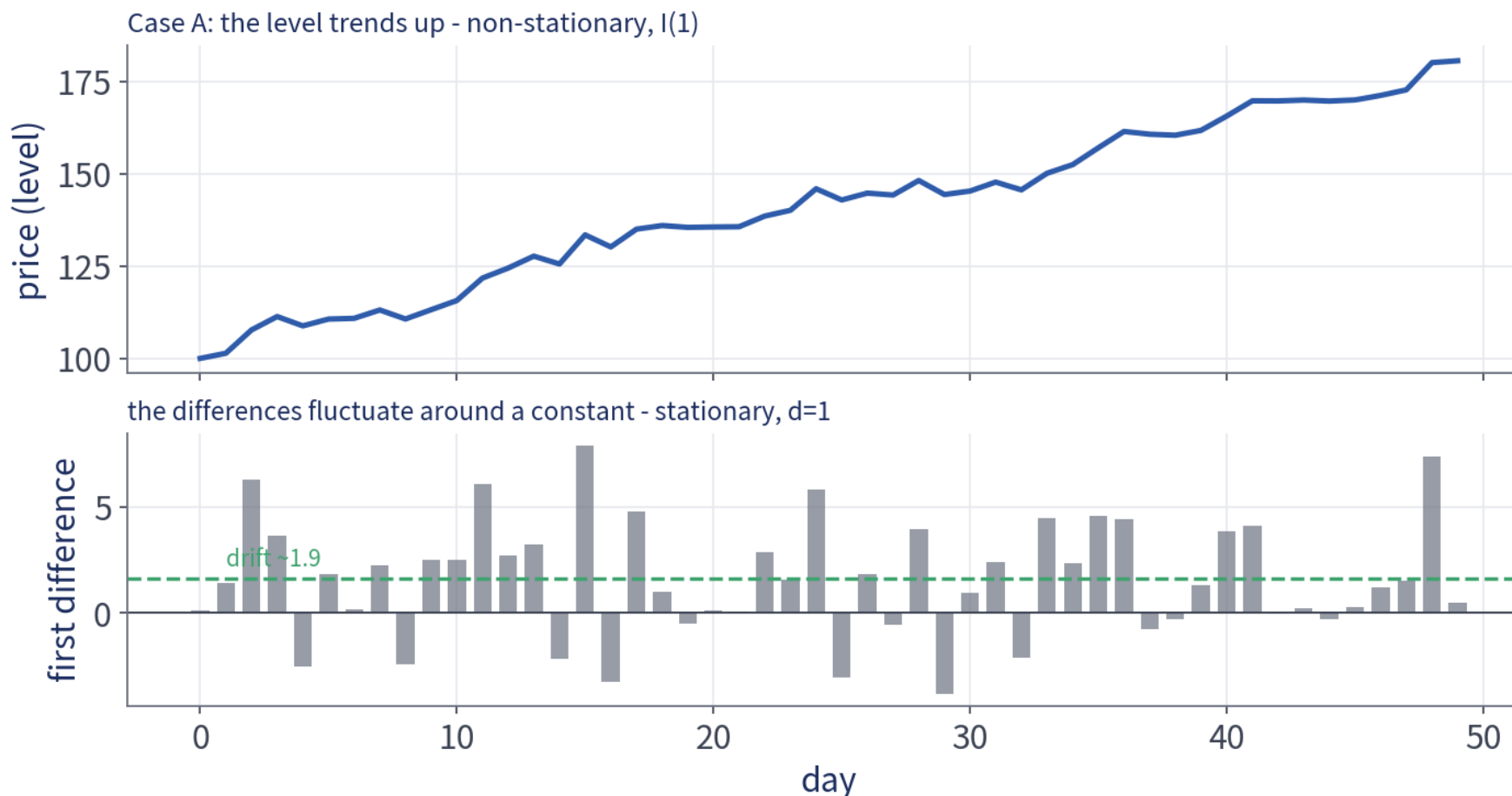
- The level fails a stationarity test.
- The differences pass it.
- So d equals one - written ARIMA(p,1,q).

Difference vs detrend

- Difference for a stochastic trend.
- Detrend - regress on time - for a deterministic one.
- ARIMA is for the stochastic kind.

The Trend Vanishes in the Differences

The top panel shows the price wandering steadily upward - non-stationary. The bottom panel shows its first differences, fluctuating around a constant drift with no trend at all. One differencing operation has stripped away the stochastic trend, leaving a stationary series to model



Why Difference at All

Prices versus returns

The problem with levels

- Prices trend, so ARMA can't model them.
- Their mean and variance drift over time.
- Regressions on them can be spurious.

The differencing fix

- The changes are stationary - ARMA applies.
- In finance, those changes are returns.
- One difference unlocks the whole toolkit.

Identify and Forecast

Model the change, then integrate back

Once differenced, the series is just ARMA - so we identify the order and forecast. The differences here look like a constant drift, making the price a random walk with drift: ARIMA(0,1,0) with a trend. We forecast the differences, then integrate them back into a level forecast that continues the trend, with bands that widen as the

The identification

- Difference once - d is one.
- The differences are a constant drift.
- So the model is a random walk with drift.

What we'll do by hand

- Forecast the drift forward.
- Cumulate it onto the last price.
- Widen the bands with the horizon.

ARIMA and Its Forecast

Difference, model, integrate back

$\text{ARIMA}(p, d, q)$: difference d times, then model as $\text{ARMA}(p, q)$; random walk = $\text{ARIMA}(0, 1, 0)$

The three integers

- p autoregressive, d differences, q moving-average.
- Difference d times to reach stationarity.
- Then model the rest as ARMA.

The forecast

- Forecast the differenced series first.
- Then cumulate to recover the level.
- The trend carries forward.

Integrate the Forecast Back

Add the drift, step by step

$$\hat{y}_{T+h} = y_T + \sum_{i=1}^h \widehat{\Delta y} = y_T + h\hat{\mu} : \quad 113 + 1(1.86) = \mathbf{114.9}, \quad 113 + 2(1.86) = \mathbf{116.7}$$

The mechanics

- Start from the last price, 113.
- Add the drift of 1.86 each step.
- That cumulates the differenced forecast.

The path

- 114.9, then 116.7, then 118.6.
- A straight-line continuation.
- The trend persists, it doesn't revert.

The Widening Bands

Uncertainty grows with the horizon

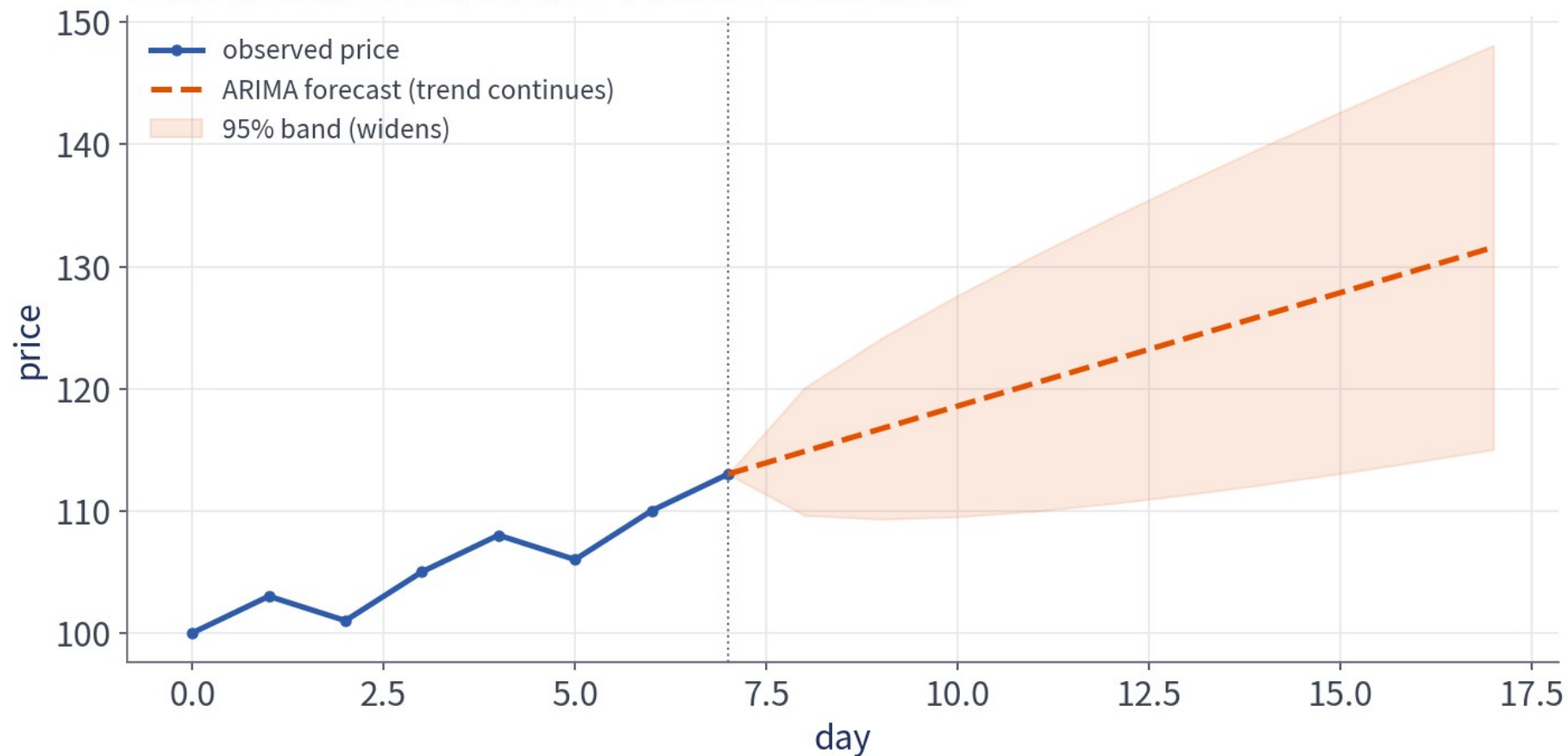
95% band = $\pm 1.96 \sigma \sqrt{h}$: ± 5.2 ($h=1$) $\rightarrow$ ± 9.1 ($h=3$) (trend continues, uncertainty grows)

Because the forecast integrates independent differences, its variance grows with the horizon - the band widens with the square root of h , from about plus-or-minus 5.2 at one step to 9.1 at three. The further out, the less certain the level.

The Trend Carries Forward

The observed prices climb, and the ARIMA forecast extends that climb in a straight line - the stochastic trend continues rather than reverting to any mean. The shaded band fans out with the horizon, because each step adds another independent uncertain difference to the running total

Case B: the stochastic trend continues - it does not revert to a mean



Trends Continue, Not Revert

The signature of a stochastic trend

What ARIMA forecasts

- A trend that continues indefinitely.
- Uncertainty that grows without bound.
- The behavior of a random walk with drift.

What a stationary model would do

- Revert to a fixed long-run mean.
- Bands that settle to a constant width.
- The wrong picture for a trending price.

The ARIMA Workflow

Difference, check model, integrate

The routine is four steps. Difference the series until it's stationary, using tests to fix the order d . Model the stationary differences as ARMA, identifying p and q . Forecast those differences. Then integrate back to recover the level, so the trend continues. One operation - differencing - bridges a trending price and the entire

The differencing (Case A)

- Prices 100 to 113, differences around 1.9.
- Stationary after one difference.
- Order d is one.

ry

The forecast (Case B)

- Drift forward: 114.9, 116.7, 118.6.
- Bands widen with the square root of h .
- The trend continues.

Reproduce the Differencing and the Forecast

Direct Claude Code to check your arithmetic

GOAL Difference a price series, confirm the order, then forecast by integrating back.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - differencing:

- * for prices `[100,103,101,105,108,106,110,113]`, print the first differences and their mean;
- * run `adf` on the level and the differences.

CASE B - forecast:

- * treat it as a random walk with drift;
- * print the 1-, 2-, 3-step level forecasts;
- * print the 95% bands (widening with `sqrt(h)`).

Then plot the level and differences, and the forecast with its band.

Check against the slides

- differences `+3,-2,+4,...`; drift 1.86.
- level non-stationary, diffs stationary.
- forecast 114.9, 116.7, 118.6.
- bands `+/-5.2 -> +/-9.1`.

Deliverable: `part1_practice.py` whose printed numbers match the differences and the forecast path.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the 'I' in ARIMA stand for?

Integration - the series is differenced d times

Q2 How do you pick the order d ?

Difference until stationarity tests pass

Q3 How does an ARIMA level forecast behave?

The trend continues; it doesn't revert to a mean

All three come from the differencing, the order of integration, and the integrated forecast.

Three Pitfalls to Avoid

Where ARIMA trips people

Common mistakes

- "Difference until it looks flat, just in case."
- "Detrending and differencing are the same."
- "An ARIMA forecast reverts to a mean."

The reality

- Over-differencing adds noise - test for the right d .
- Difference stochastic trends, detrend deterministic ones.
- The stochastic trend continues, with widening bands.

Two Calculations to Remember

Part 1 applications in one screen

1

One difference makes it stationary

The prices climbed from 100 to 113, but their first differences - +3, -2, +4, and so on - fluctuated around a drift of 1.86 with no trend. The level was non-stationary; the differences were stationary; the order of integration is one.

2

Integrate the forecast back

Treating the price as a random walk with drift, the level forecast added 1.86 each step - 114.9, 116.7, 118.6 - continuing the trend rather than reverting. The 95% band widened with the square root of the horizon, from 5.2 to 9.1.

3

Differencing bridges trend and toolkit

Difference until stationary, model the changes as ARMA, forecast, then integrate back. That single operation lets the full stationary machinery handle a trending price.

Next — Part 2 practice: seasonality and SARIMA, differencing away the yearly and weekly cycles too.

★ WEEK 4 · PART 2 of 3

Seasonality and SARIMA

Modeling the Repeating Rhythm

Seasonal differencing · $SARIMA(p,d,q)(P,D,Q)_s$ · the airline model

Part 1 tamed the trend. Part 2 adds the season — the yearly, quarterly, or weekly pattern the airline data showed — by applying the very same differencing and ARMA ideas at the seasonal lag.

2

PART 2 · SEASONALITY

Adding the Season

What seasonality is, how a seasonal difference removes it, and how SARIMA extends ARIMA with a second set of orders at the seasonal lag.

One Pattern Left to Capture

Part 1 removed the trend; the season remains

1

Differencing killed the trend, not the season

A regular difference removes a slow drift, but a yearly pattern survives — it repeats every s periods, and ordinary ARIMA cannot see it.

2

Seasonality is just memory at a longer lag

Today looks like the same month last year. That is autocorrelation at lag s — the same idea as AR/MA, moved out to the seasonal lag.

3

So we reuse everything, at lag s

Seasonal differencing, seasonal AR, seasonal MA: the Week-3 and Part-1 tools, applied a second time at the season. That is SARIMA.

SARIMA in one line: do ARIMA twice — once at lag 1, once at the seasonal lag s .

What Is Seasonality?

A pattern that repeats on a fixed calendar cycle

Seasonality is a systematic pattern that recurs every fixed number of periods s — 12 for monthly data, 4 for quarterly, 7 for daily-with-weekly rhythm.

How to spot it

- The same shape repeats every s periods, year after year.
- Peaks and troughs land in the same months each cycle.
- The ACF shows spikes at lags s , $2s$, $3s$, ...

Why it matters

- Ignoring it leaves obvious, predictable structure unmodeled.
- It shows up in residuals as leftover seasonal spikes.
- Capturing it sharply improves short-horizon forecasts.

Additive vs Multiplicative Seasonality

Do the swings stay constant, or grow with the level?

Additive

- The seasonal swing is a fixed size each cycle.
- $\text{Observed} = \text{trend} + \text{season} + \text{noise}$.
- The peaks and troughs stay the same height over time.

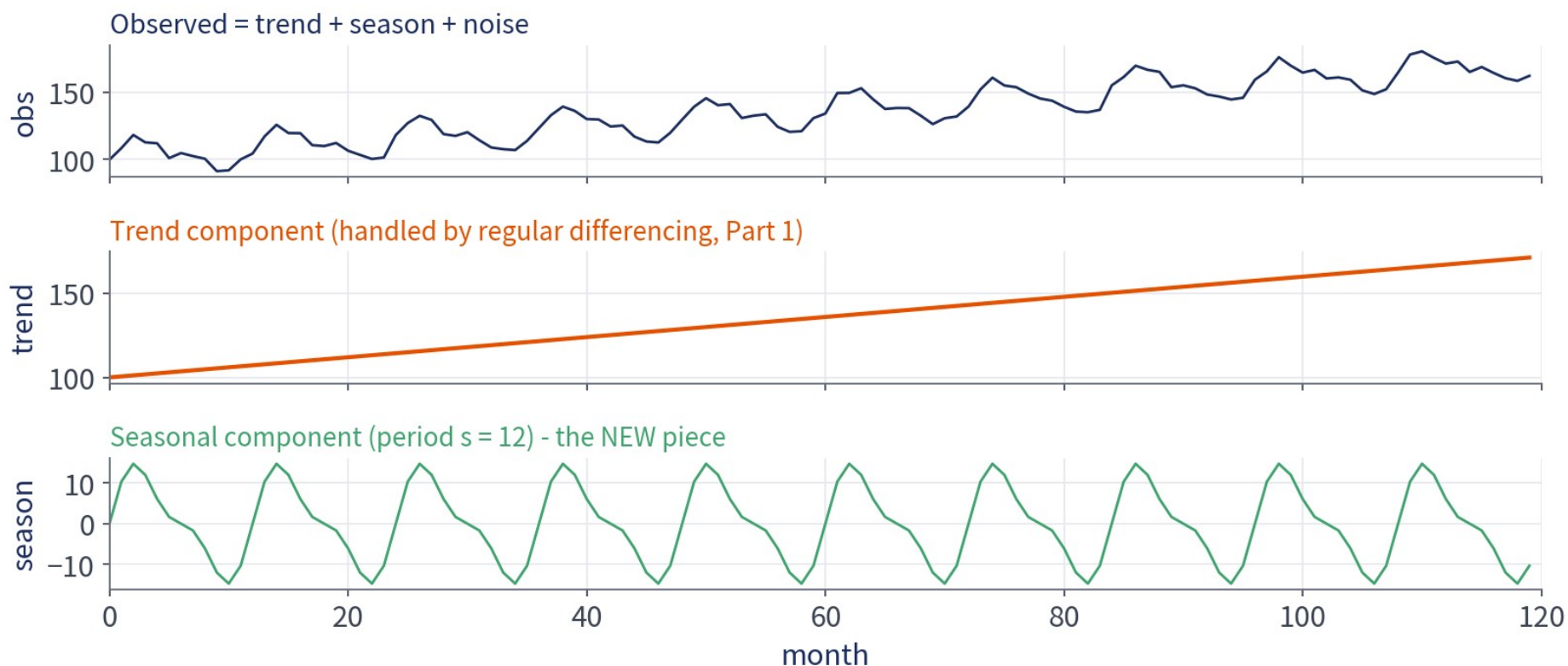
Multiplicative

- The swing scales with the level — bigger as the series grows.
- $\text{Observed} = \text{trend} \times \text{season} \times \text{noise}$ (the airline data).
- Take logs to turn it back into the additive case.

The Anatomy of a Seasonal Series

Any seasonal series splits into a trend (handled by regular differencing, Part 1), a repeating seasonal component (the new piece), and noise.

The anatomy of a seasonal series



Why Plain ARIMA Misses the Season

Adjacent-lag memory cannot reach the seasonal lag

An ordinary ARIMA models correlations at nearby lags (1, 2, 3...). A yearly pattern lives at lag 12 — too far away for a low-order model to capture without help.

The symptom

- Fit an ARIMA and the residual ACF still spikes at lag s .
- The model explains the wiggles but not the yearly rhythm.
- You could raise the order to 12+, but that is wasteful.

The fix

- Add terms that act directly at the seasonal lag s .
- A seasonal difference removes the repeating pattern outright.
- Seasonal AR/MA terms mop up what remains — enter SARIMA.

Seasonal Differencing

Subtract the value from one season ago

$$\nabla_s X_t = (1 - L^s) X_t = X_t - X_{t-s} \quad (\text{seasonal difference})$$

What it does

- Replaces each value with its change from s periods earlier.
- For monthly data, $(1 - L^{12})$ subtracts the same month last year.
- One pass removes a stable, repeating seasonal pattern.

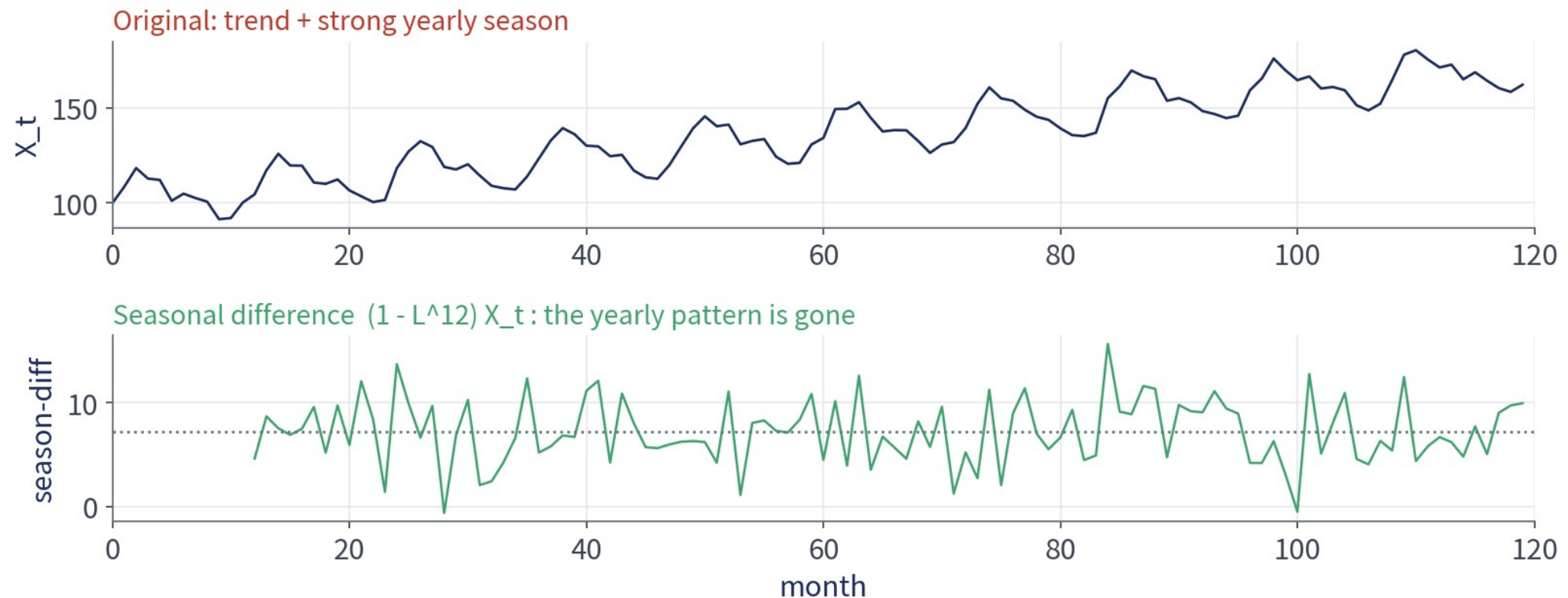
Regular vs seasonal

- A regular difference $(1 - L)$ targets the trend.
- A seasonal difference $(1 - L^s)$ targets the season.
- Trending, seasonal data often needs BOTH.

Seasonal Differencing in Action

Top: a series with a strong yearly pattern. Bottom: after $(1 - L^{12})$, the repeating season is gone and the series fluctuates around a flat mean.

Seasonal differencing subtracts the value from one season ago



The Seasonal ACF/PACF Signature

Read the seasonal lags exactly as you read the near ones

Seasonality announces itself as spikes at the seasonal lags — s , $2s$, $3s$ — in the ACF and PACF. You identify the seasonal orders (P , Q) from those spikes, just like p and q .

Where to look

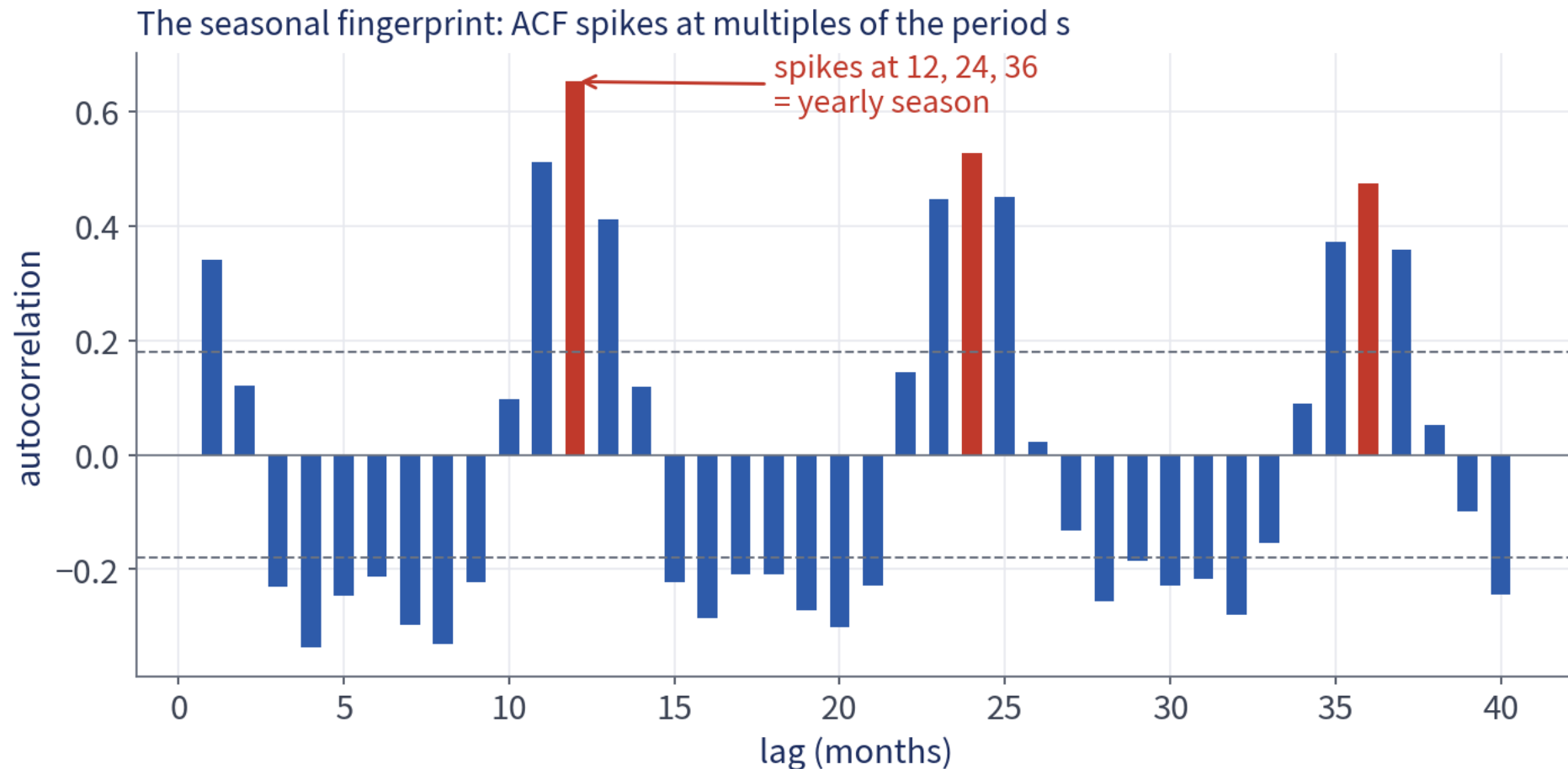
- Scan the ACF/PACF at lags s , $2s$, $3s$ specifically.
- A cut-off at the seasonal lags names the seasonal order.
- Slow decay across seasonal lags means you need a seasonal difference.

The two-region rule

- Near lags ($1, 2, 3\dots$) → the non-seasonal (p, d, q).
- Season lags ($s, 2s\dots$) → the seasonal (P, D, Q).
- Read the two regions independently.

The Seasonal Fingerprint

The ACF spikes at multiples of the period s (here 12, 24, 36). Those seasonal spikes are what tell you a seasonal term — or a seasonal difference — is needed.



SARIMA(p,d,q)(P,D,Q)_s

Two sets of orders, one at each timescale

$$\underbrace{\phi(L) \Phi(L^s)}_{\text{AR} \times \text{seasonal AR}} (1-L)^d (1-L^s)^D X_t = \underbrace{\theta(L) \Theta(L^s)}_{\text{MA} \times \text{seasonal MA}} \varepsilon_t$$

The two halves

- (p, d, q): the ordinary ARIMA at adjacent lags (Part 1).
- (P, D, Q): the SAME structure applied at the seasonal lag s.
- s is the period — 12 monthly, 4 quarterly, 7 daily.

Why it is elegant

- No new theory — the seasonal part mirrors the non-seasonal part.
- The two multiply together in one clean equation.
- Everything you learned in Weeks 3–4 transfers directly.

SARIMA, Decoded

Six integers: (p, d, q) handle the short-run dynamics and trend; (P, D, Q) handle the season at lag s . Same ideas, applied twice.

SARIMA, Decoded

Six integers: the first three handle ordinary dynamics, the last three handle the season.

(p, d, q) (P, D, Q) ^{s}
period

Non-seasonal part (p, d, q)

p - AR order on adjacent lags
 d - regular differences (the trend)
 q - MA order on adjacent shocks

Exactly the ARIMA of Part 1 -
short-run dynamics and the trend.

Seasonal part (P, D, Q) at lag s

P - AR order on seasonal lags $(s, 2s, \dots)$
 D - seasonal differences $(1 - L^s)$
 Q - MA order on seasonal lags

The same ideas, applied at the
season lag s (e.g. $s = 12$ for monthly).

The Seasonal Polynomials

AR and MA, written at the seasonal lag

$$\Phi(L^s) = 1 - \Phi_1 L^s - \dots - \Phi_P L^{Ps}, \quad \Theta(L^s) = 1 + \Theta_1 L^s + \dots + \Theta_Q L^{Qs}$$

Reading them

- $\Phi(L^s)$ is a seasonal AR polynomial in the lag operator L^s .
- $\Theta(L^s)$ is the seasonal MA counterpart.
- They act only at lags that are multiples of s .

The parallel

- These mirror $\phi(L)$ and $\theta(L)$ from the non-seasonal part.
- P and Q are the seasonal AR and MA orders.
- The full model is the product of both polynomials.

The Airline Model

The most famous SARIMA of all

$$(1 - L)(1 - L^{12}) X_t = (1 + \theta L)(1 + \Theta L^{12}) \varepsilon_t \quad \text{ARIMA}(0, 1, 1)(0, 1, 1)_{12}$$

What it is

- ARIMA(0,1,1)(0,1,1)_12 — one regular and one seasonal difference.
- Plus one MA term at lag 1 and one at lag 12.
- Just two parameters, yet it fits the airline data beautifully.

Why it endures

- It captures trend AND season with remarkable parsimony.
- It is the default starting point for monthly seasonal data.
- A perfect illustration of 'simple but useful' (Box).

Identifying the Seasonal Orders

A disciplined reading of the two lag regions

The routine

- First difference (d) and seasonally difference (D) as needed.
- Read the near lags for (p, q) ; read lags $s, 2s$ for (P, Q) .
- Keep all four orders small — parsimony still rules.

Practical tips

- Start from the airline model and adjust from its residuals.
- A single seasonal difference usually suffices ($D = 1$).
- Confirm with AIC/BIC and residual diagnostics, as always.

The Seasonal Cheat Sheet

Read each lag region separately

What you see	Near lags (1, 2, 3...)	Seasonal lags (s, 2s, 3s...)
ACF cuts off	→ MA order q	→ seasonal MA order Q
PACF cuts off	→ AR order p	→ seasonal AR order P
Slow decay	→ difference d	→ seasonal difference D

The whole trick: apply the Week-3 cheat sheet twice — once near lag 1, once around the season lag s.

Combining Regular and Seasonal Differences

How d and D work together

Trending, seasonal data often needs one of each: a regular difference for the trend and a seasonal difference for the pattern — applied together as $(1 - L)(1 - L^s)$.

The usual recipe

- $d = 1$ removes the trend; $D = 1$ removes the season.
- Together they flatten both at once.
- Higher d or D is rarely needed and risks over-differencing.

How to decide

- Difference for the trend first, then check for seasonal spikes.
- Add a seasonal difference only if those spikes persist.
- Stop as soon as the ACF is clean at both regions.

Building a SARIMA

The ARIMA loop, with a seasonal pass

1 · Difference

d and D

Regular difference for trend; seasonal for the pattern.

2 · Identify

both regions

Read near lags for (p,q); season lags for (P,Q).

3 · Estimate

fit + rank

Fit SARIMA; compare candidates by AIC/BIC.

4 · Diagnose

residuals

Check no seasonal spikes remain in the residuals.

5 · Forecast

season + trend

Predict; the forecast repeats the season.

Same five steps as Part 1 — you simply do the differencing and identification at two lags.

Forecasting With SARIMA

The forecast carries both the trend and the season

A SARIMA forecast reproduces everything it learned: the trend keeps rising and the seasonal shape repeats, cycle after cycle, with intervals that widen as the horizon grows.

What you see

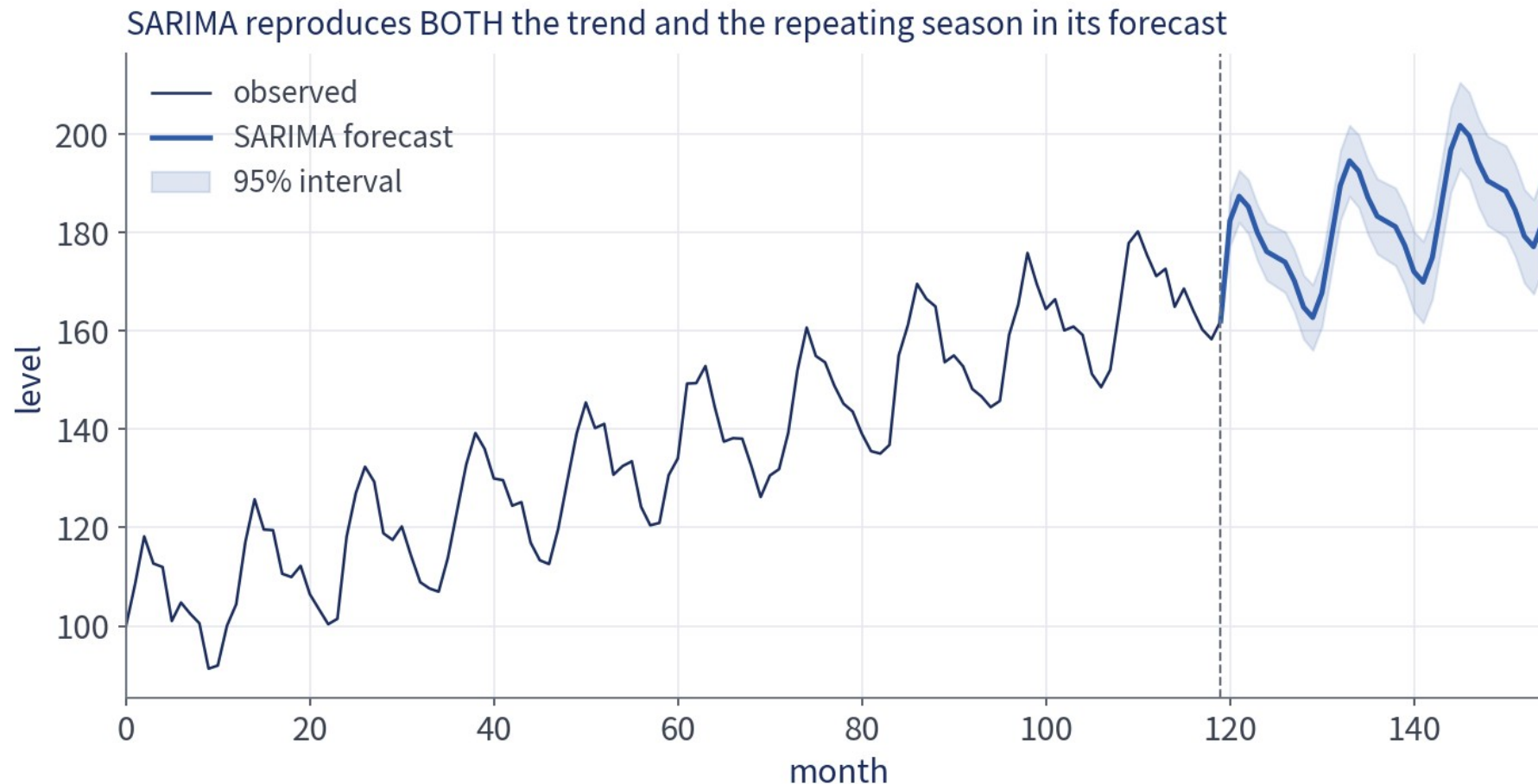
- The point forecast continues the trend (from the 'I').
- It overlays the repeating seasonal shape (from the 'S').
- The interval fans out, widest at the far horizon.

Why it is powerful

- One model captures level, trend, and season together.
- Short-horizon seasonal forecasts can be strikingly accurate.
- It remains a strong baseline for demand and macro data.

A SARIMA Forecast Repeats the Season

The forecast continues the upward trend while faithfully reproducing the yearly shape — trend from the 'I', season from the 'S', uncertainty from the fanning interval.



Seasonality in Finance

Powerful for some series, treacherous for returns

Where it genuinely helps

- Energy and commodity demand (heating in winter).
- Retail sales and volumes (holiday peaks).
- Quarterly macro series — GDP, earnings, inventories.

Where to be skeptical

- Raw return series show little robust seasonality.
- Calendar 'anomalies' are weak and often vanish out of sample (Week 1).
- Model seasonality on volumes and fundamentals, rarely on returns.

Fit a SARIMA to Seasonal Data

Direct Claude Code through the seasonal workflow

GOAL Take a clearly seasonal series, choose d and D , identify all six orders, and forecast the repeating pattern.

[INPUT] Prompt · Claude Code

Write and run `sarima_lab.py` that:

1. loads a monthly seasonal series (e.g. the classic airline-passengers data, $s = 12$);
2. plots it and its ACF/PACF, marking the spikes at lags 12, 24, 36;
3. applies a regular and a seasonal difference, then fits `SARIMA(0,1,1)(0,1,1)[12]` and one alternative, comparing AIC/BIC;
4. forecasts 24 months ahead with 95% bands and plots the level.

Explain which orders you chose and why.

How to work with it

- Look for ACF spikes exactly at 12, 24, 36.
- The forecast should repeat the yearly shape.
- Check residuals for leftover seasonal spikes.
- Keep it; Part 3 automates the whole search.

Deliverable: `sarima_lab.py` + a 24-month seasonal forecast with intervals.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

Monthly data still spikes at lags 12 and 24 after one regular difference. What next?

Seasonal difference

Q2

In $\text{SARIMA}(p,d,q)(P,D,Q)_s$, which orders act at the seasonal lag?

(P, D, Q)

Q3

You fit $\text{ARIMA}(0,1,1)(0,1,1)_{12}$. What kind of data is this classic for?

**Monthly,
trend+season**

All three come straight from the two-region reading and the seasonal difference.

Three Pitfalls to Avoid

Where seasonal modeling goes wrong

Common mistakes

- "Raise the ARIMA order until it captures the season."
- "Seasonal spikes in the residuals are fine."
- "Returns must have a tradable seasonal pattern."

The reality

- Use a seasonal term at lag s — do not brute-force a high order.
- Leftover seasonal spikes mean the model is incomplete — revise.
- Return seasonality is weak and unstable; be skeptical (Week 1).

Three Things to Carry Forward

Part 2 in one screen

1

Seasonality is memory at the lag s

A pattern that repeats every s periods shows up as ACF spikes at $s, 2s, 3s$ — the same autocorrelation idea, moved out to the seasonal lag.

2

SARIMA is ARIMA applied twice

(p, d, q) at adjacent lags and (P, D, Q) at the seasonal lag s . A seasonal difference removes the pattern; seasonal AR/MA terms mop up the rest.

3

The airline model is your default

$\text{ARIMA}(0,1,1)(0,1,1)_{12}$ captures trend and season with just two parameters — the parsimonious starting point for monthly seasonal data.

Next — Part 3: building and forecasting ARIMA/SARIMA in practice, with automation and evaluation.

★ WEEK 4 · PART 2 · PRACTICE

Seasonality and SARIMA, Worked by Hand

Difference Away the Calendar, Then Decode the Six Orders

Seasonal differencing, the airline model, and combining d with D

Part 2's tools, with the arithmetic shown. We seasonally difference a quarterly series so the repeating pattern cancels, decode the six-integer SARIMA notation and the famous airline model, then combine a regular and a seasonal difference to make a trending, seasonal series stationary.

2P

PART 2 · APPLICATIONS

Two Timescales, Two Differences

Case A: seasonally difference a quarterly series - subtract the value from one season ago - and watch the repeating pattern vanish. Case B: decode SARIMA and the airline model, then combine a seasonal and a regular difference to reach stationarity. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

Seasonality repeats on a fixed cycle

A pattern that recurs every s periods - a quarter, a month, a week. Plain ARIMA's near-lag memory can't reach it.

2

Seasonal differencing subtracts a season ago

Replace each value with its change from s periods back. The repeating pattern cancels, leaving the underlying movement.

3

SARIMA has two sets of orders

Six integers: regular ones for short-run dynamics and trend, seasonal ones for the cycle at lag s . The airline model is the classic.

Everything here is a calculator exercise: one seasonal difference, one decoding, one combined difference.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a seasonal series.
- Seasonally difference, or combine two differences.
- Then read what's left.

What you should watch

- How subtracting a season ago cancels the cycle.
- How the six SARIMA orders split the work.
- How d and D together reach stationarity.

Seasonal Differencing

Subtraction season ago

A quarterly series peaks every fourth quarter - a strong, repeating season a plain difference can't remove. Seasonal differencing does: subtract the value from exactly one season, four quarters, ago. If the season repeats, those subtractions cancel it, leaving only the year-over-year change. We compute it and watch the pattern

The setup

- Two years of quarterly data.
- A Q4 peak and Q3 dip each year.
- Period s is four.

What we'll do by hand

- Subtract each quarter from the same quarter last year.
- See the seasonal swing cancel.
- Read the leftover growth.

Seasonal Differencing

One season back, not one step

$$\Delta_s y_t = y_t - y_{t-s} \quad (\text{subtract the value from one season ago; here } s=4 \text{ quarters})$$

Why not a plain difference

- A regular difference reaches only one step back.
- It can't see the pattern s periods away.
- The season would survive it.

What seasonal differencing does

- It reaches back a full season.
- Same quarter, same month, last cycle.
- The repeating pattern cancels.

Subtract Last Year's Quarter

Same quarter, one year back

	Q1	Q2	Q3	Q4
Year 1	100	110	105	140
Year 2	108	118	113	148
Delta-4	+8	+8	+8	+8

Subtracting each quarter from the same quarter a year earlier gives +8 every time. The Q4 peak minus last year's Q4 peak, the Q3 dip minus last year's dip - the season cancels exactly, leaving only the growth.

The Season Cancels

A constant is what's left

$$\Delta_4 y = 108 - 100, 118 - 110, 113 - 105, 148 - 140 = 8, 8, 8, 8 \text{ (the season cancels)}$$

What happened

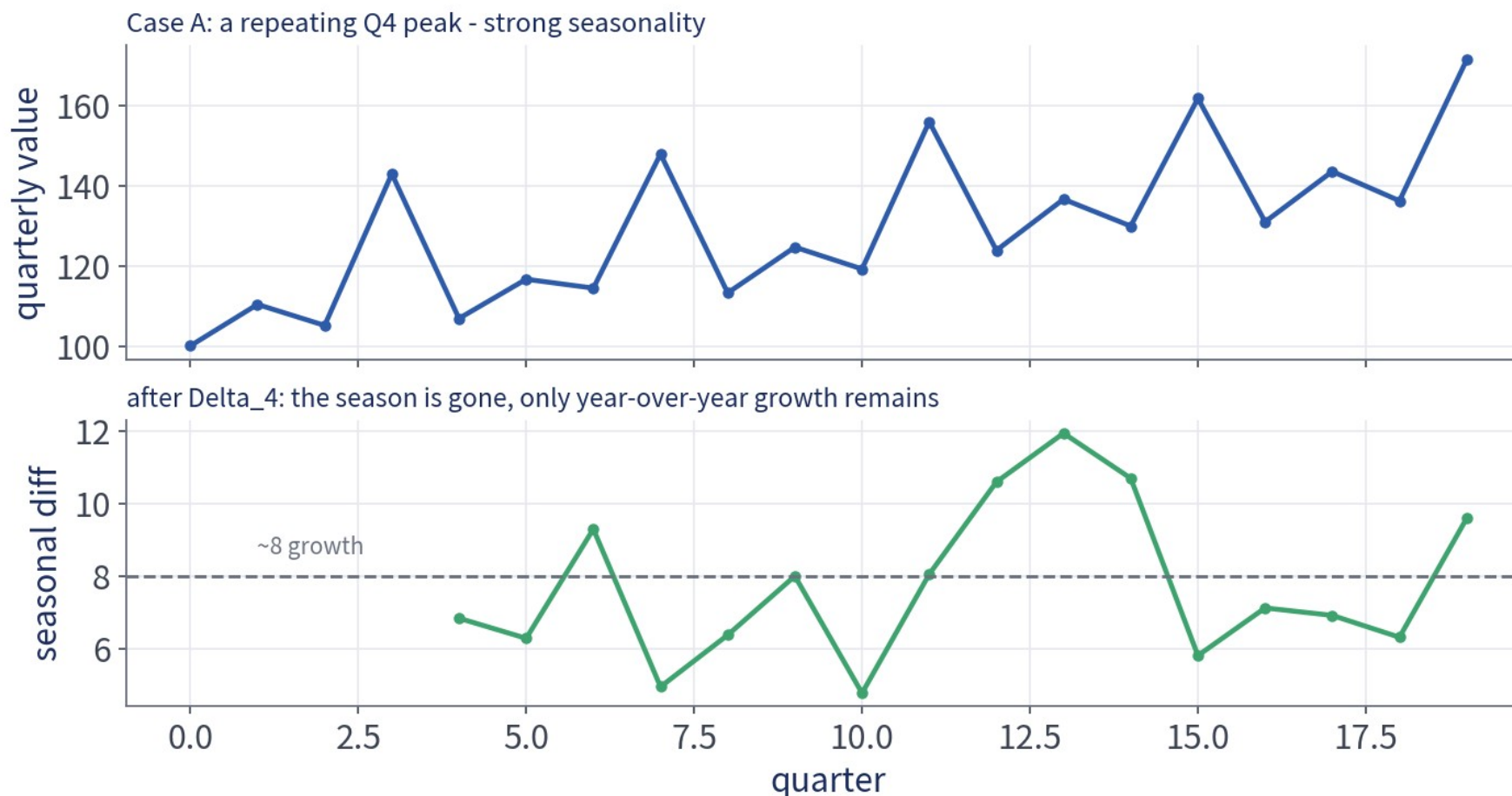
- Every seasonal difference is the same 8.
- The repeating swing has vanished.
- Only the year-over-year growth remains.

What it means

- The seasonal difference is stationary here.
- So the seasonal order D is one.
- No leftover trend, so no regular difference needed.

The Season Disappears

The top panel shows the raw series, spiking every fourth quarter - an unmistakable seasonal pattern. The bottom panel shows the seasonal difference, which flattens to a near-constant around the year-over-year growth. Subtracting one season ago has erased the calendar pattern entirely.



Additive or Multiplicative

Which season, which difference

Additive seasonality

- The swings stay a constant size.
- A seasonal difference removes them cleanly.
- That's the case we just worked.

Multiplicative seasonality

- The swings grow with the level.
- Take logs first, then difference.
- Logs turn the growing swing into a constant one.

The Airline Model

Six orders and two differences combined

When a series has both a trend and a season, one difference isn't enough. SARIMA carries six orders - regular ones for the short run and trend, seasonal ones for the cycle. The airline model is the classic. We decode it, then combine a seasonal and a regular difference on a series that has both, watching each difference

The two jobs

- The seasonal difference removes the cycle.
- The regular difference removes the trend.
- Together they reach stationarity.

What we'll do by hand

- Decode the airline model's six orders.
- Seasonally difference a trending series.
- Then difference again to flatten it.

SARIMA, Decoded

Two sets of orders, one at each scale

$\text{SARIMA}(p, d, q)(P, D, Q)_s$: (p, d, q) short-run & trend, $(P, D, Q)_s$ the season at lag s

The regular orders

- Lower-case p, d, q .
- Short-run autoregression, differencing, and MA.
- Exactly the ARIMA you already know.

The seasonal orders

- Upper-case P, D, Q at lag s .
- The same three ideas, one season apart.
- They handle the calendar cycle.

The Airline Model

The most famous SARIMA

airline model: $\text{SARIMA}(0, 1, 1)(0, 1, 1)_{12}$ – $d=1$ trend, $D=1$ season, $q=Q=1$ MA terms

Reading the orders

- d is one: a regular difference for the trend.
- D is one: a seasonal difference for the cycle.
- q and Q are one: an MA at each scale.

Why it endures

- Two differences and two MA terms.
- Fits a huge range of seasonal series.
- Named for the airline-passenger data it modeled.

Combining d and D

Season first, then trend

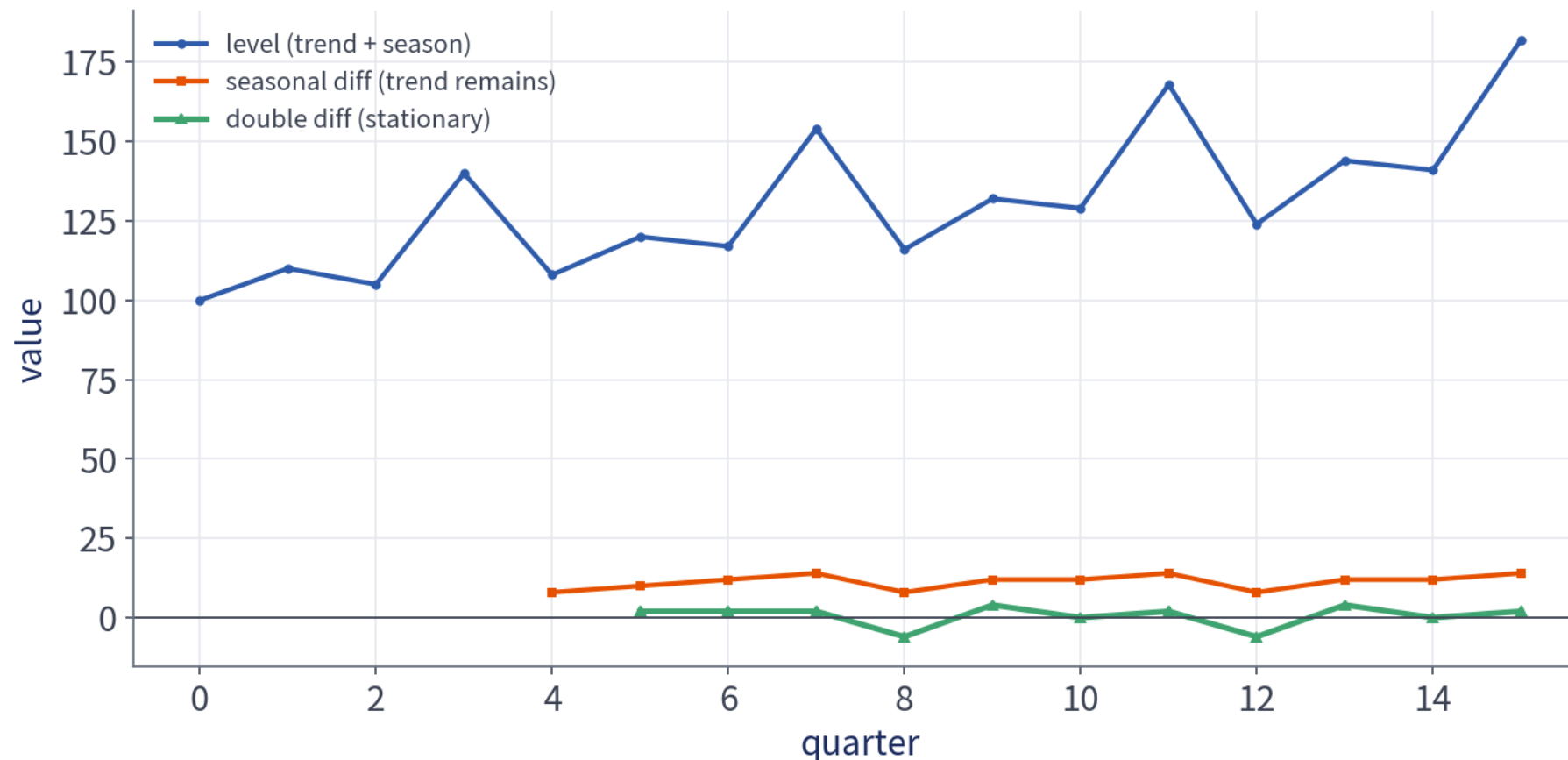
$$\Delta_4 y = 8, 10, 12, 14 \xrightarrow{\Delta} 2, 2, 2 \quad (D=1 \text{ removes season, } d=1 \text{ removes leftover trend})$$

On a series with both, the seasonal difference gives 8, 10, 12, 14 - the season is gone, but a trend remains. A regular difference of that yields a constant 2. Both differences together - D and d each one - make it stationary.

Two Differences to Flatness

The blue level carries both a trend and a season. The orange seasonal difference removes the cycle but still slopes upward. The green double difference - seasonal then regular - lands flat around a constant: stationary at last. Each difference does one job and both are needed.

Case B: seasonal then regular differencing -> a flat, stationary series



Seasonality in Finance

Powerful, but treacherous for returns

Where it helps

- Volume, earnings, and macro series cycle.
- SARIMA models those calendar effects well.
- Forecasts respect the known season.

Where it misleads

- Return seasonality is mostly noise or arbitrated away.
- Fitting a season to returns overfits.
- Use it for flows and fundamentals, not prices.

The SARIMA Workflow

Seasonal trend model for trend.

The routine extends ARIMA to two timescales. Seasonally difference to remove the calendar cycle, and regularly difference to remove the trend - using tests to fix D and d . Model what remains as a seasonal ARMA, identifying the regular and seasonal orders. Then forecast and integrate both differences back. The airline model is

The season (Case A)

- Delta-4 gave a constant 8.
- The Q4 peak cancelled exactly.
- Seasonal order D is one.

The combination (Case B)

- Seasonal diff 8,10,12,14 - trend remains.
- Regular diff of that: constant 2.
- Airline model: d and D both one.

Reproduce the Differences and the Model

Direct Claude Code to check your arithmetic

GOAL Seasonally difference, combine with a regular difference, and fit a SARIMA.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that:  
CASE A - seasonal difference:  
* for quarterly [100,110,105,140,108,118,113,148],  
  print the seasonal difference (lag 4).  
CASE B - combine and fit:  
* for [100,110,105,140,108,120,117,154], print the  
  seasonal diff, then the regular diff of that;  
* fit SARIMAX with order (0,1,1) and seasonal  
  (0,1,1,4) and print the summary.  
Then plot the level, seasonal diff, and double diff.
```

Check against the slides

- Case A seasonal diff: all 8.
- Case B seasonal diff: 8,10,12,14.
- then regular diff: 2,2,2.
- airline: (0,1,1)(0,1,1).

Deliverable: part2_practice.py whose printed numbers match the seasonal and combined differences.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does a seasonal difference subtract?

The value from one full season ago

Q2 What are the six SARIMA orders?

Regular p, d, q and seasonal P, D, Q at lag s

Q3 When do you need both d and D ?

When the series has both a trend and a season

All three come from the seasonal difference, the SARIMA notation, and the combined differencing.

Three Pitfalls to Avoid

Where SARIMA trips people

Common mistakes

- "A regular difference removes the season too."
- "Difference seasonally as many times as it takes."
- "Fit a season to stock returns."

The reality

- Only a seasonal difference reaches the cycle at lag s .
- Usually D of one suffices - over-differencing adds noise.
- Return seasonality is mostly noise - use flows instead.

Two Calculations to Remember

Part 2 applications in one screen

1

Seasonal differencing cancels the cycle

Subtracting each quarter from the same quarter a year earlier gave a constant +8 - the Q4 peak and Q3 dip cancelled exactly. One seasonal difference removed the repeating pattern, leaving only year-over-year growth.

2

Combine d and D for trend plus season

On a series with both, the seasonal difference gave 8, 10, 12, 14 - season gone, trend left - and a regular difference of that gave a constant 2. The airline model, SARIMA(0,1,1)(0,1,1)₁₂, encodes exactly this: one difference at each scale.

3

Two timescales, one framework

Seasonal differencing for the calendar, regular differencing for the trend, then a seasonal ARMA on what remains. Powerful for volume, earnings, and macro series - but treacherous for returns.

Next — Part 3 practice: the full ARIMA modeling routine, from identification through diagnostics on real data.

★ WEEK 4 · PART 3 of 3

ARIMA in Practice

Building, Automating, and Honestly Evaluating Forecasts

Transforms · auto_arima · rolling back-tests · the finance caveat

Theory meets reality. Part 3 is the practitioner's playbook: prepare the data, let automation search, validate out of sample, and know exactly where ARIMA belongs — and where it does not.

3

PART 3 · IN PRACTICE

From a Model to a Real Forecast

Data preparation, automated model selection, rigorous out-of-sample evaluation, and a clear map of where ARIMA/SARIMA earns its keep in finance.

Everything, Put to Work

Parts 1–2 built the model; Part 3 uses it well

1

We can specify any ARIMA or SARIMA

Trend via d , season via D , dynamics via (p, q) and (P, Q) . The specification problem is solved — now comes the craft of applying it.

2

Real forecasting is mostly discipline

Prepare the data, let tools search the orders, and — above all — validate on data the model never saw. That discipline is what separates a forecast from a guess.

3

And know the boundaries

ARIMA shines on trending, seasonal, memory-bearing series; it is nearly useless on raw returns. Knowing which is the whole point of this course.

Part 3 in one line: prepare, automate the search, validate out of sample, and respect the limits.

The Complete Pipeline

Weeks 2–4 collapse into one routine

Transform, difference to stationarity, select the orders, diagnose the residuals, then forecast and back-test. Every week of this block is one station on that line.

The stations

- Stationarize: difference for trend (d) and season (D).
- Identify: ACF/PACF or an automated search over orders.
- Diagnose: confirm the residuals are white noise.

The discipline that binds it

- Prefer the simplest model that passes diagnostics.
- Never trust in-sample fit — validate out of sample.
- Compare every forecast against a naive benchmark.

Transform Before You Model

Stabilize the variance first

$$y_t^{(\lambda)} = \begin{cases} \frac{y_t^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \ln y_t & \lambda = 0 \end{cases} \quad (\text{Box-Cox})$$

When and why

- If seasonal swings grow with the level, take logs first.
- A log turns multiplicative seasonality into additive.
- Box-Cox generalizes the log with a tunable parameter λ .

What it buys

- A roughly constant variance — what ARIMA assumes.
- More symmetric, better-behaved residuals.
- Remember to invert the transform when reporting forecasts.

Outliers and Missing Data

Clean inputs, or clean nothing

Outliers

- A single spike can distort every estimated coefficient.
- Flag it, understand its cause, and consider an intervention term.
- Never delete data silently — document every adjustment.

Missing values & gaps

- ARIMA needs an evenly-spaced, complete series.
- Interpolate short gaps; flag long ones as structural breaks.
- Align the calendar carefully — especially across markets.

Manual vs Automated Selection

Two ways to choose the orders

Hand-reading the ACF/PACF builds intuition; automated search scales it. In practice you let the tool propose, then apply human judgment to the shortlist.

Manual (Box-Jenkins)

- Read the ACF/PACF and propose orders yourself.
- Slow, but it teaches you what the data is doing.
- Essential for learning — and for sanity-checking the tool.

Automated (`auto_arima`)

- Searches many $(p,d,q)(P,D,Q)$ combinations quickly.
- Ranks them by AIC or BIC and returns the best.
- Fast and consistent — but blind to context.

How auto_arima Searches

Automation, demystified

auto_arima runs unit-root tests to pick d and D , then steps through candidate orders, fitting each and keeping the one with the lowest information criterion.

What it automates

- Choosing d and D via ADF/KPSS-type tests.
- A stepwise search over (p, q) and (P, Q) .
- Ranking candidates by AIC or BIC.

What you still steer

- The transform, the season length s , and the search range.
- Whether to allow a trend/drift term.
- The final call, after reading the residuals.

What Automation Cannot Do

The tool proposes; you dispose

Its blind spots

- It optimizes a criterion, not economic sense.
- It can miss structural breaks and one-off events.
- A low AIC does not guarantee good forecasts.

Your irreplaceable role

- Judge whether the model is plausible for the data.
- Read the residual diagnostics it will not read for you.
- Validate out of sample — the criterion is only in-sample.

Comparing Candidate Models

The comparison that actually decides

$$\text{choose model} = \arg \min_{\text{candidates}} \text{out-of-sample error (not in-sample fit)}$$

The right criterion

- AIC/BIC narrow the field quickly and cheaply.
- But the final judge is out-of-sample forecast error.
- A model that fits the past can still forecast poorly.

How to compare fairly

- Score all candidates on the SAME held-out period.
- Include the naive benchmark in the comparison.
- Prefer the simpler model when scores are close.

Forecast Evaluation, Done Right

The out-of-sample rule, made concrete

Split time in order: fit on the past, forecast the future, and score only on data the model never saw. Anything else measures memorization, not forecasting.

The cardinal rule

- Never let future information leak into the fit.
- Respect time order — no shuffling, no peeking ahead.
- Score on the held-out period alone.

Why it is non-negotiable

- In-sample error always flatters complex models.
- Only out-of-sample error predicts real performance.
- This is where most forecasting projects quietly fail.

Train/Test and Rolling-Origin Back-Tests

One split is a start; rolling is rigorous

A single split

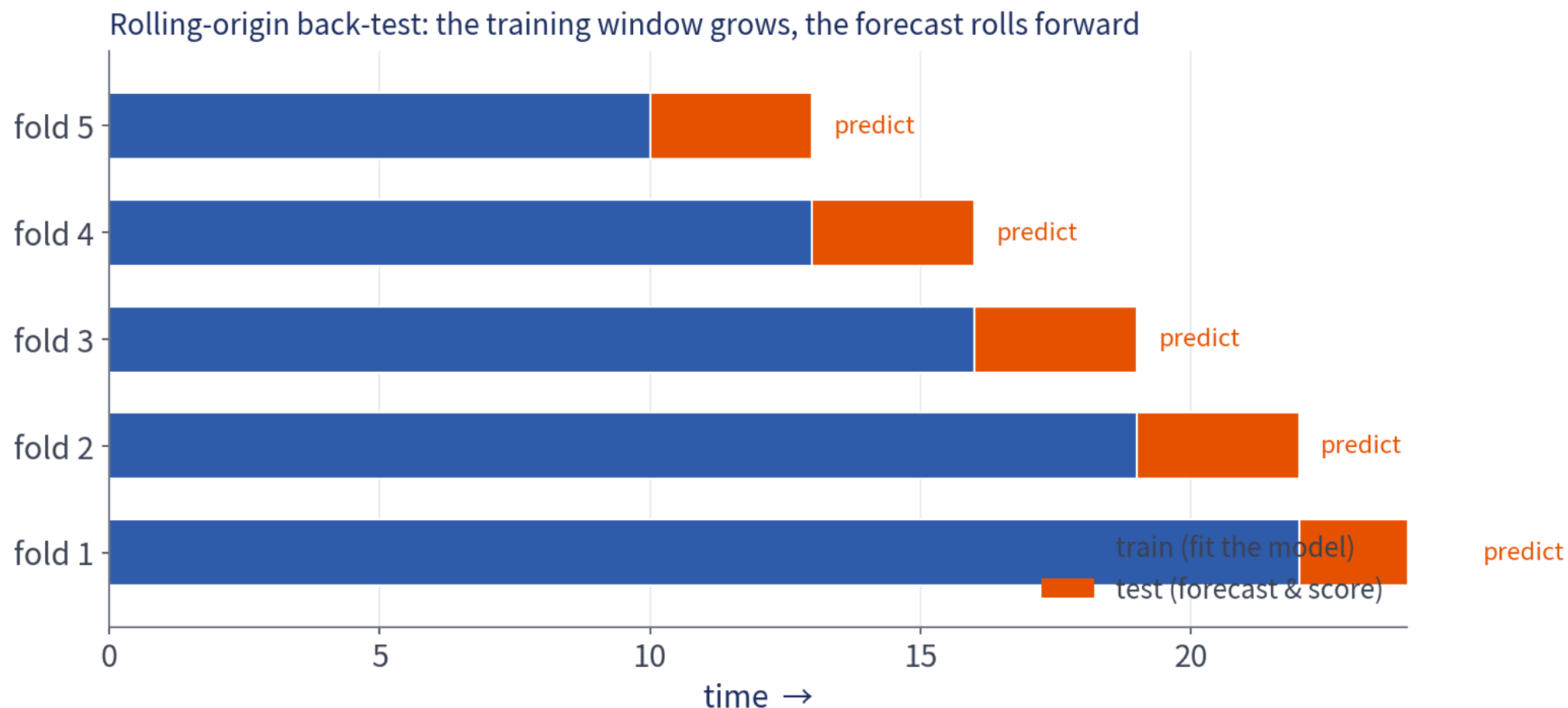
- Fit on the first part, forecast the last part, score it.
- Simple, fast, and far better than in-sample alone.
- But it rests on just one window — possibly lucky or unlucky.

Rolling-origin (walk-forward)

- Repeat the split, sliding the origin forward each time.
- Average the errors across many forecasts.
- A robust estimate of true out-of-sample accuracy.

The Rolling-Origin Back-Test

Fit on the blue window, forecast the orange, score it — then roll the origin forward and repeat. Averaging over folds gives an honest accuracy estimate.



Metrics and the Benchmark

Numbers only mean something in comparison

Element	What it is	Why it matters
RMSE / MAE	average size of the forecast error	the headline accuracy number
Naive benchmark	'no-change' or seasonal-naive forecast	the bar any model must clear
Skill score	improvement over the benchmark	turns raw error into 'is it worth it?'

A small RMSE is meaningless in isolation — always report it beside the naive benchmark's error.

Prediction Intervals and Coverage

Are your 95% bands really 95%?

$$\text{coverage} = \Pr(y_{T+h} \in [\hat{y}_{T+h} \pm z \sigma_h]) \stackrel{?}{\approx} 95\%$$

The honest question

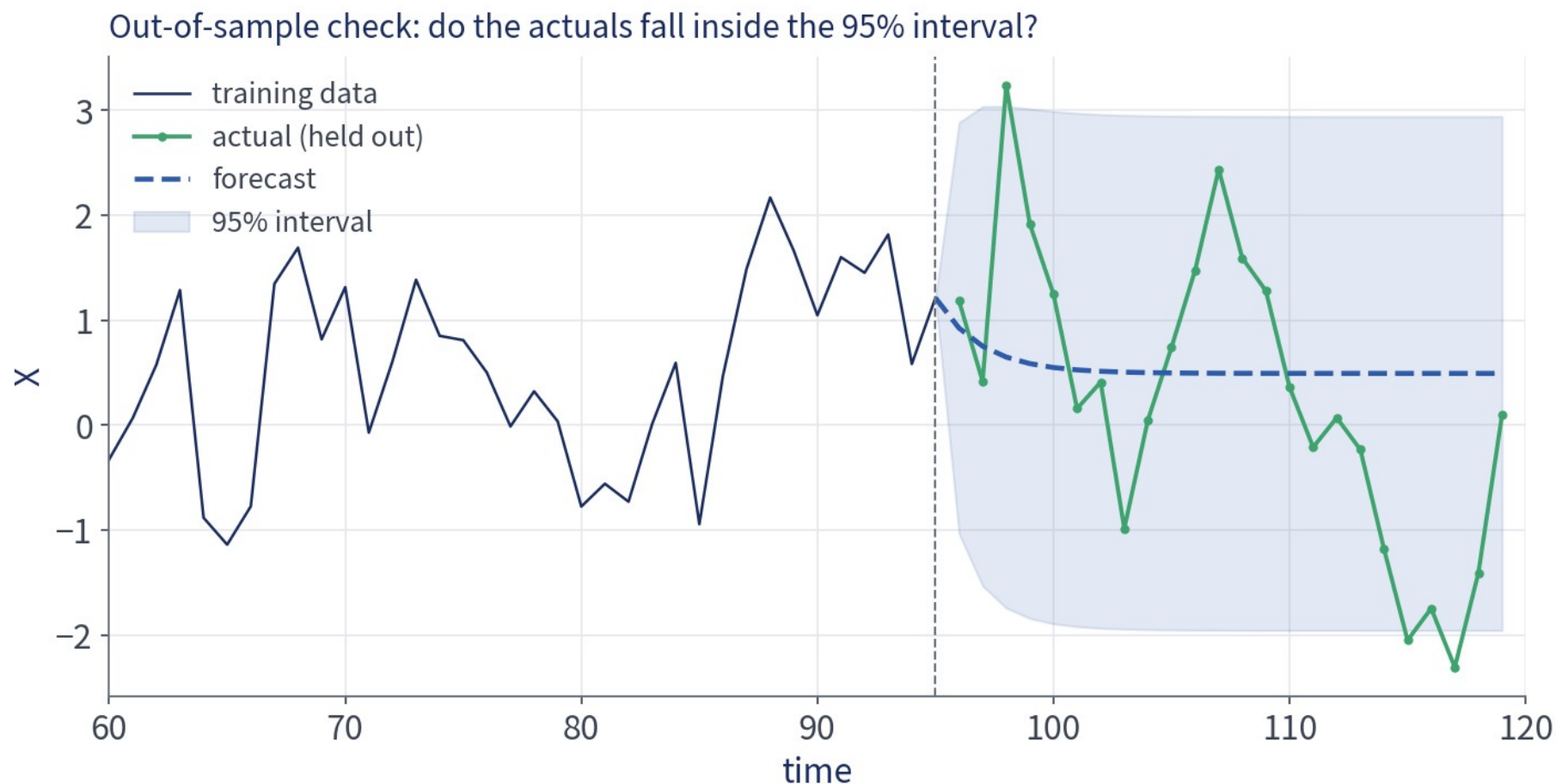
- Over many forecasts, do ~95% of actuals land in the band?
- If far fewer do, your intervals are too narrow.
- Fat-tailed or clustered errors are the usual culprit.

Why it matters most in finance

- Risk decisions hinge on the interval, not the point.
- Under-covered intervals understate tail risk badly.
- Check coverage on the back-test, not just the point error.

Out-of-Sample: Forecast vs Actual

The real test: hold out the recent data, forecast it, and see whether the actuals (green) fall inside the 95% interval. Most should — if the model is honest.



ARIMA on Prices vs Returns

The single most important practical distinction

An ARIMA on a price LEVEL is essentially a random walk with drift — useful for the trend. An ARIMA on RETURNS finds almost nothing, because returns are close to unpredictable.

Price levels ($I(1)$)

- ARIMA models the trend; the forecast continues upward.
- It is effectively a random walk with drift.
- Fine for level projection, weak for timing.

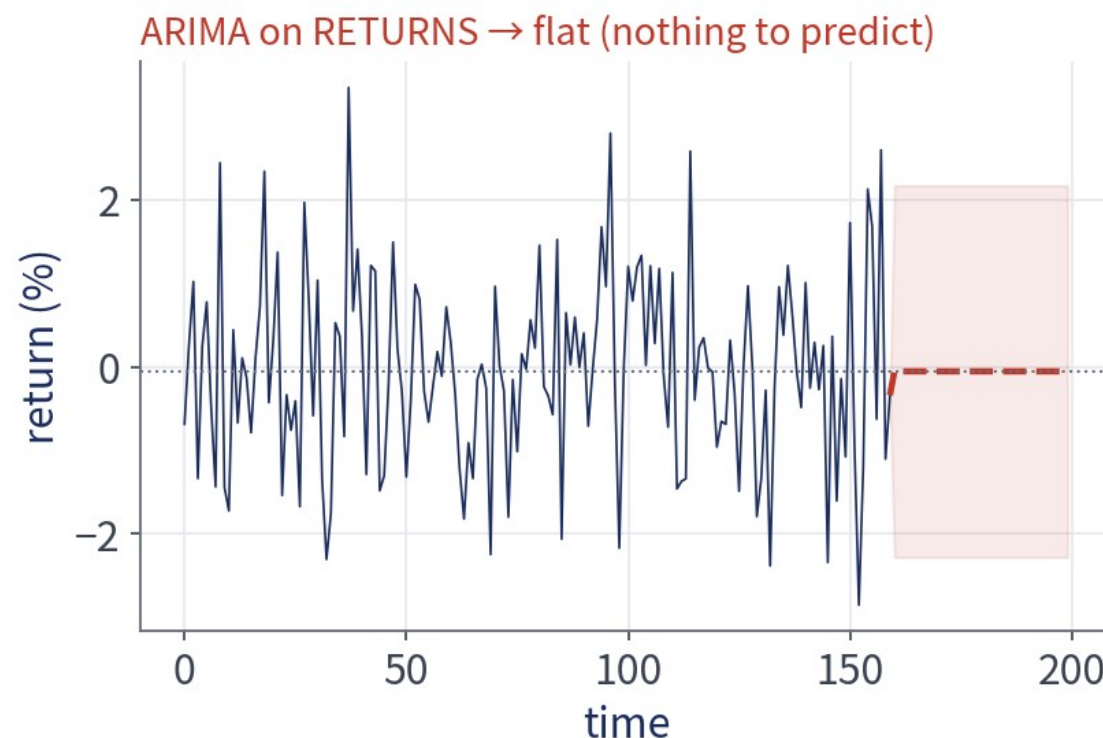
Returns ($\approx I(0)$, $\approx$ white noise)

- The ACF is near zero — no linear structure to fit.
- The forecast collapses to the mean return.
- The action is in the VARIANCE, not the mean.

Why You Model the Level, Not the Returns

Left: an ARIMA on the price level extends the trend. Right: on returns it forecasts a flat mean — there is simply no linear structure to project.

The finance caveat: model the price level, not the (near-random) returns



Where ARIMA/SARIMA Genuinely Applies

A quick fit-for-purpose guide

Domain	Good fit?	Why
Macro series (GDP, CPI)	Yes	trend + memory + clear seasonality
Demand, sales, volumes	Yes	strong, stable seasonal patterns
Asset price levels	Partly	$I(1)$; ARIMA $\approx$ random walk with drift
Asset returns	No	$\approx$ unpredictable; model the volatility instead

Rule of thumb: model levels, macro, and demand; be deeply skeptical of forecasting returns.

When to Move Beyond ARIMA

Three signals the tool has run out of road

The warning signs

- The residual MEAN is clean but its VARIANCE clusters.
- The relationship shifts — a structural break or regime change.
- The dynamics are clearly nonlinear or threshold-driven.

What comes next

- Clustered variance → volatility models (ARCH/GARCH).
- Regime shifts → Markov-switching or time-varying models.
- Nonlinearity → threshold, state-space, or ML approaches.

The Bridge to Volatility Models

Where the returns' structure was hiding

ARIMA models the mean and finds returns unpredictable. But their squared or absolute values are strongly autocorrelated — volatility clusters. That is the doorway to GARCH.

The handoff

- ARIMA/SARIMA: model the level, trend, and season.
- Whatever mean structure exists, it captures.
- What it leaves behind is time-varying variance.

The same tools, one level down

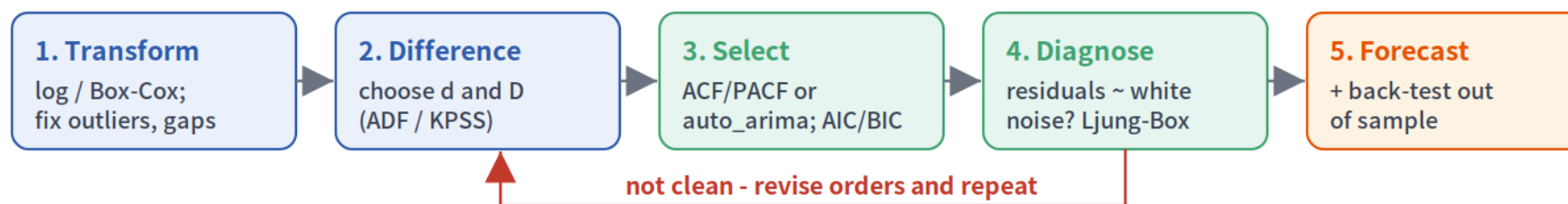
- Apply the ACF logic to squared returns.
- Model the variance as its own autoregressive process.
- That is exactly what GARCH does — the natural sequel.

The Practitioner Pipeline

Transform, difference, select, diagnose, forecast — with a feedback loop when residuals aren't clean, and an off-ramp to volatility models when the mean has nothing left to give.

The ARIMA/SARIMA Practitioner Pipeline

Weeks 2 to 4 as one routine - with an off-ramp to volatility models when the mean has nothing left.



ARIMA / SARIMA is right when...

- the series has a trend and/or a season
- there is genuine linear memory in the mean
- macro, demand, volumes, fundamentals
- residuals become clean white noise

Move beyond it when...

- the mean is ~unpredictable (raw returns)
- residual VARIANCE clusters (volatility)
- relationships shift (regime change)

then the next step is a volatility model (GARCH).

An Automated, Back-Tested Pipeline

Direct Claude Code from raw data to an honest score

GOAL Build an end-to-end pipeline: transform, auto-select the ARIMA/SARIMA, then back-test it against a naive benchmark.

[INPUT] Prompt · Claude Code

Write and run `pipeline_lab.py` that:

1. loads a trending, seasonal series and log-transforms it if the swings grow;
 2. runs `auto_arima` to pick $(p,d,q)(P,D,Q)[s]$;
 3. back-tests with a rolling origin, reporting RMSE for the model AND a seasonal-naive benchmark, plus interval coverage;
 4. plots one out-of-sample forecast vs actual.
- Tell me: does the model beat the benchmark, and are the 95% intervals well-calibrated?

How to work with it

- Sanity-check `auto_arima`'s pick against the ACF.
- Beating a seasonal-naive benchmark is the real win.
- Coverage near 95% means honest intervals.
- Then try it on returns — watch it add nothing.

Deliverable: `pipeline_lab.py` + a back-test report with benchmark and coverage.

Three Questions Before You Finish

If you can answer these, Week 4 has landed

Q1

auto_arima returns the lowest-AIC model. Is that automatically your best forecaster?

No — validate OOS

Q2

Your 95% intervals contain only 70% of actuals in the back-test. What's wrong?

Intervals too narrow

Q3

Should you fit an ARIMA to a stock's daily returns to time trades?

No — returns ~ noise

All three separate a textbook fit from a forecast you could actually trust.

Three Pitfalls to Avoid

Where practical forecasting goes wrong

Common mistakes

- "auto_arima chose it, so it must be right."
- "A great back-test RMSE means a great model."
- "ARIMA on returns will give me a trading signal."

The reality

- Automation optimizes a criterion, not judgment — always sanity-check.
- Compare to a benchmark and check interval coverage, not just RMSE.
- Returns are $\approx$ unpredictable; the edge, if any, is in volatility.

Week 4 in Three Ideas

Pulling the whole week together

1

ARIMA integrates the trend; SARIMA adds the season

Part 1: difference d times, then ARMA. Part 2: do it again at the seasonal lag s . The airline model captures both with just two parameters.

2

Practice is preparation, automation, and honest evaluation

Part 3: transform and clean the data, let `auto_arima` search, then validate with rolling back-tests against a naive benchmark — and check interval coverage.

3

Know the boundary: mean vs variance

ARIMA models the mean and finds little in raw returns. The structure that remains lives in the variance — the doorway to volatility models.

Next — modeling what ARIMA leaves behind: volatility clustering, with ARCH and GARCH.

★ WEEK 4 · PART 3 · PRACTICE

ARIMA in Practice, Worked by Hand

Select the Order, Then Prove It Out of Sample

AIC and BIC, RMSE against a naive benchmark, and coverage

Part 3's full routine, with the arithmetic shown. We score candidate ARIMA orders with AIC and BIC, then evaluate the chosen model out of sample - RMSE against a naive forecast, interval coverage, and the crucial distinction between modeling prices and returns.

3P

PART 3 · APPLICATIONS

Fit, Then Prove It Holds Up

Case A: score candidate ARIMA orders with AIC and BIC, and see them trade fit against parsimony. Case B: evaluate out of sample - RMSE versus a naive forecast, prediction-interval coverage, and why prices and returns need different scrutiny. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

Selection balances fit and complexity

AIC and BIC both reward likelihood and punish parameters - BIC more harshly. The lowest score wins, and they can disagree.

2

Only out-of-sample results count

In-sample fit always improves with parameters. The honest test is forecast error on data the model never saw.

3

Prices and returns need different scrutiny

On price levels an ARIMA forecast collapses to the naive one. The real edge, if any, shows up on returns - out of sample.

Everything here is a calculator exercise: one AIC/BIC table, one out-of-sample RMSE, one coverage count.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from candidate models or a test set.
- Score them, or measure forecast error.
- Then decide what to trust.

What you should watch

- How AIC and BIC penalize parameters differently.
- How RMSE compares to a naive benchmark.
- Why prices flatter a model and returns test it.

Selecting the Order

Fit against parsimony

With several plausible ARIMA orders, we need a principled choice. AIC and BIC both start from the model's likelihood and subtract a penalty for each parameter - BIC's penalty grows with the sample size, so it leans toward simpler models. We score four candidates and watch the two criteria nick differently

The candidates

- From a random walk to ARIMA(2,1,1).
- Each with its log-likelihood.
- Two hundred observations.

What we'll do by hand

- Compute AIC for each candidate.
- Compute BIC for each.
- See which order each criterion picks.

AIC and BIC

Reward fit, punish parameters

$$\text{AIC} = -2 \ln L + 2k, \quad \text{BIC} = -2 \ln L + k \ln T \quad (\text{fit } vs \text{ parameter count } k)$$

The shared part

- Both start from minus-two log-likelihood.
- Lower means a better fit.
- Both add a penalty for parameters.

The difference

- AIC's penalty is a flat two per parameter.
- BIC's is the log of the sample size.
- So BIC punishes complexity harder.

Score the Candidates

Four orders, two criteria

order	logL	k	AIC	BIC
(0,1,0)	-305	1	612.0	615.3
(1,1,0)	-298	2	600.0	606.6
(1,1,1)	-296	3	598.0	607.9
(2,1,1)	-295.5	4	599.0	612.2

Each AIC is minus-two times the log-likelihood plus twice the parameter count; each BIC swaps in the log of 200, about 5.3, per parameter. Reading down the last two columns reveals where each criterion bottoms out.

Where They Disagree

AIC richer, BIC simpler

AIC $\rightarrow (1, 1, 1)$ (598.0), BIC $\rightarrow (1, 1, 0)$ (606.6) (BIC's $\ln T$ penalty prefers parsimony)

AIC's pick

- Lowest at ARIMA(1,1,1), 598.0.
- It tolerates the extra MA term.
- Because its penalty is mild.

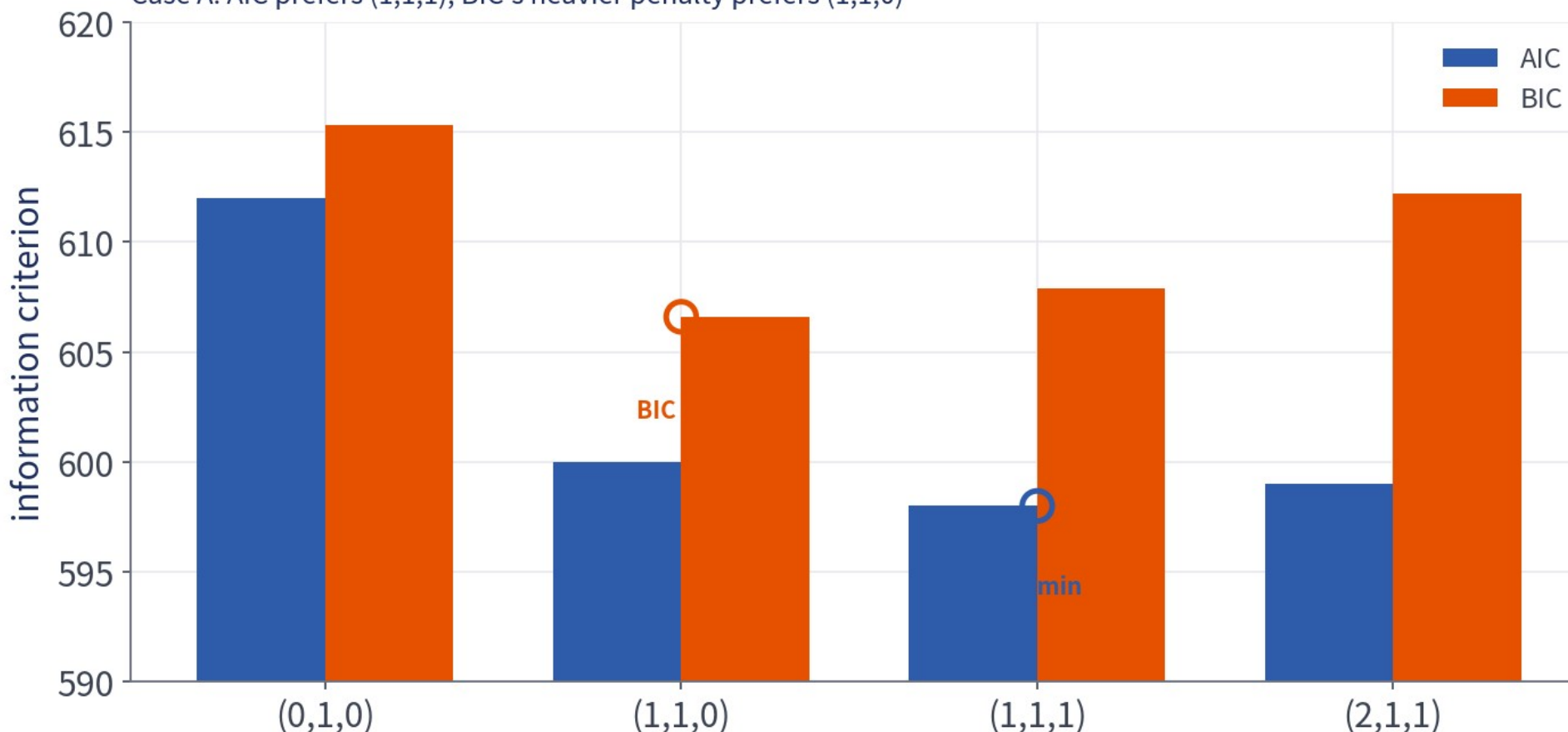
BIC's pick

- Lowest at ARIMA(1,1,0), 606.6.
- The extra term isn't worth the penalty.
- So it chooses the simpler model.

Two Criteria, Two Choices

AIC (blue) bottoms out at ARIMA(1,1,1), rewarding the slightly richer fit. BIC (orange), with its heavier per-parameter penalty, bottoms out one model earlier at (1,1,0). Both rise again for the over-parameterized (2,1,1). When they disagree, the simpler BIC choice usually forecasts more reliably.

Case A: AIC prefers (1,1,1); BIC's heavier penalty prefers (1,1,0)



Manual or Automated

Both land in the same place

Automated search

- Tools like `auto_arima` grid over orders.
- They minimize AIC or BIC for you.
- Fast, and a sensible default.

Manual judgment

- Reading ACF and PACF suggests orders too.
- Prefer the simpler model when scores are close.
- Automation is a starting point, not the last word.

Proving It Out of Sample

The only test that counts

A model that fits the past can still forecast badly. So we hold out a test set and measure the error there - root mean squared error - against the simplest possible benchmark, the naive forecast that just repeats the last value. We compute both, check whether the prediction interval really covers, and confront the prices-versus-

The evaluation

- Five held-out test points.
- ARIMA errors versus naive errors.
- Then count interval coverage.

What we'll do by hand

- Compute RMSE for both.
- Compare the two.
- Check the 95% band and the prices caveat.

Out-of-Sample RMSE

Error on data the model never saw

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y_t - \hat{y}_t)^2} \quad (\text{on the held-out test set, never the training data})$$

The rule

- Train on the first part, test on the rest.
- Never score on the training data.
- Rolling-origin repeats this many times.

The benchmark

- Compare to the naive last-value forecast.
- Beating it is the minimum bar.
- Many models quietly fail to.

ARIMA vs Naive

The out-of-sample scoreboard

test point	1	2	3	4	5
ARIMA error	+0.3	-0.5	+0.2	+0.4	-0.3
naive error	+0.5	-0.7	+0.4	+0.6	-0.5

$$\text{ARIMA } \sqrt{0.126} = \mathbf{0.355} < \text{naive } \sqrt{0.302} = \mathbf{0.550} \quad (\approx 35\% \text{ better out-of-sample})$$

Squaring, averaging, and rooting the errors gives an ARIMA RMSE of 0.355 against the naive 0.550 - about a 35% improvement out of sample. The model earns its keep on this return series.

Coverage, and Prices vs Returns

Two final checks

on price levels: $\hat{y}_{T+1} \approx y_T$ (the naive forecast) $\Rightarrow$ no edge; test on RETURNS

Interval coverage

- The 95% band was plus-or-minus 0.7.
- All five test errors fell inside.
- So the interval is roughly calibrated.

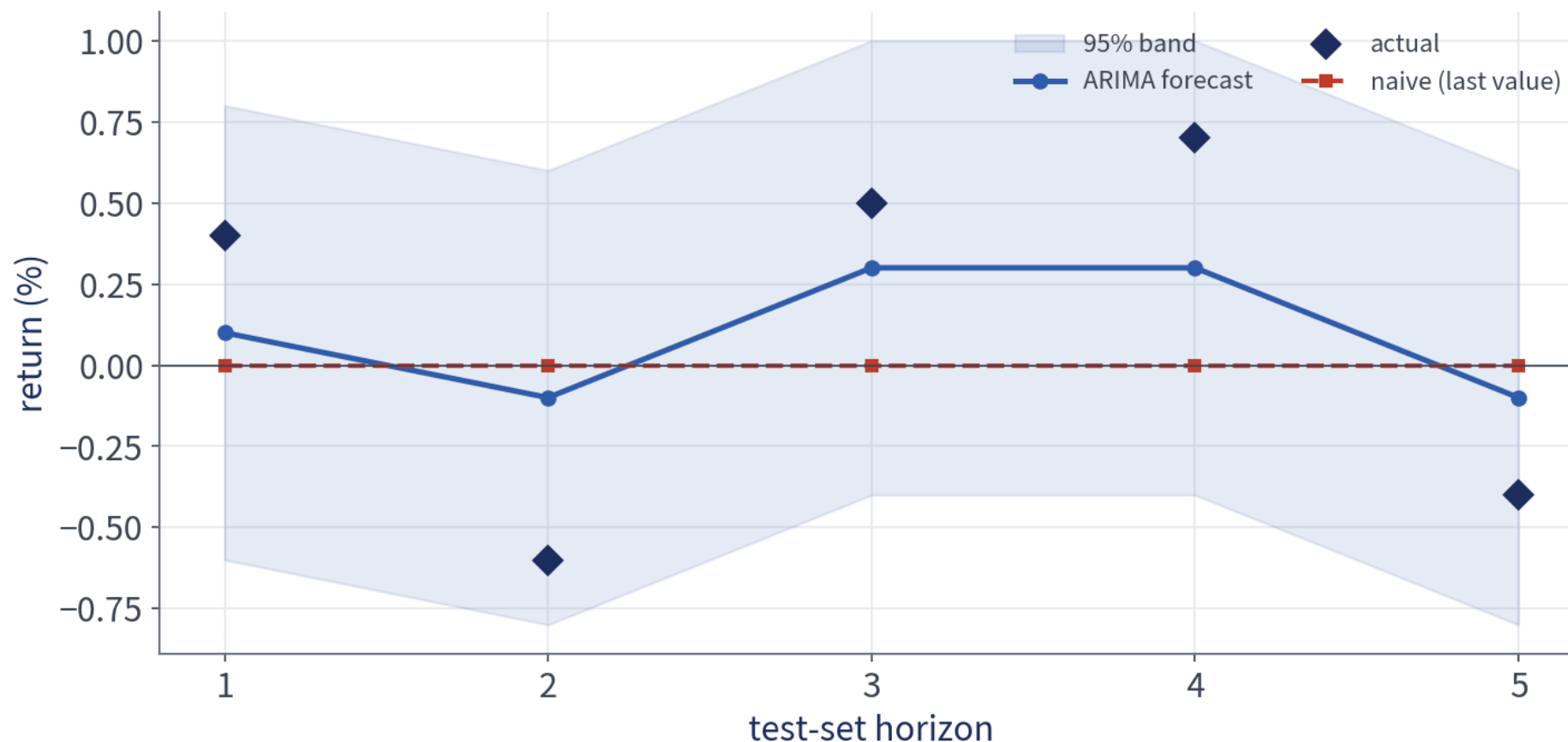
The prices caveat

- On price levels, ARIMA just echoes the last price.
- That's the naive forecast - no edge.
- The honest test is RMSE on returns.

Tracking the Test Set

The blue ARIMA forecast, with its 95% band, follows the held-out actuals (diamonds) more closely than the flat naive forecast (dashed red). Every actual falls inside the band - coverage holds - and the tighter tracking is what turns into the lower RMSE. This is the out-of-sample proof.

Case B: ARIMA tracks the test actuals better than the naive forecast



The Practical Distinction

Prices flatter, returns test

Why prices mislead

- A price is a random walk plus drift.
- The best forecast is nearly the last value.
- In-sample fit looks great but adds nothing.

Why returns are honest

- Any genuine structure lives in the returns.
- Out-of-sample RMSE reveals it - or its absence.
- Always benchmark against the naive forecast.

The ARIMA Pipeline

Weeks 2 to 4 in our course

The full routine is compact. Transform to stabilize the variance and clean the data. Difference to stationarity and identify the orders, letting AIC or BIC decide. Fit, then check residuals and score out of sample against a naive benchmark, verifying interval coverage. And always ask whether you're modeling a price or a return - the answer

The selection (Case A)

- AIC picks (1,1,1), BIC picks (1,1,0).
- Prefer the simpler when close.
- Automation as a starting point.

The evaluation (Case B)

- ARIMA RMSE 0.355 beats naive 0.550.
- Coverage holds: 5 of 5 inside.
- Test on returns, not prices.

Reproduce the Selection and the Evaluation

Direct Claude Code to check your arithmetic

GOAL Score candidates by AIC/BIC, then evaluate out of sample against a naive forecast.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - selection:  
  * with T=200 and logL for orders (0,1,0),(1,1,0),  
    (1,1,1),(2,1,1), print each AIC and BIC;  
  * report which order each criterion picks.  
CASE B - evaluation:  
  * for ARIMA errors [0.3,-0.5,0.2,0.4,-0.3] and naive  
    [0.5,-0.7,0.4,0.6,-0.5], print both RMSEs;  
  * count how many errors fall in a +/-0.7 band.  
Then plot AIC/BIC and the forecast vs actuals.
```

Check against the slides

- AIC min (1,1,1)=598.
- BIC min (1,1,0)=606.6.
- RMSE 0.355 vs 0.550.
- coverage 5/5.

Deliverable: part3_practice.py whose printed numbers match the AIC/BIC table and the RMSEs.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do AIC and BIC differ?

BIC's per-parameter penalty grows with the sample size

Q2 What's the honest test of a forecast?

Out-of-sample error versus a naive benchmark

Q3 Why test on returns, not prices?

On prices, ARIMA collapses to the naive forecast

All three come from the AIC/BIC selection, the out-of-sample RMSE, and the prices-versus-returns caveat.

Three Pitfalls to Avoid

Where ARIMA practice trips people

Common mistakes

- "Pick the model with the best in-sample fit."
- "A low AIC guarantees good forecasts."
- "Great fit on prices means a good model."

The reality

- In-sample fit always improves - score out of sample.
- Beat the naive benchmark, or you have nothing.
- Prices flatter any model; test on returns.

Two Calculations to Remember

Part 3 applications in one screen

1

Score the orders, mind the penalty

Across four candidates, AIC bottomed at ARIMA(1,1,1) at 598 while BIC, with its log-of-T penalty, bottomed one model earlier at (1,1,0). When the two criteria disagree, the simpler BIC choice usually forecasts more reliably.

2

Prove it out of sample

On five held-out points, the ARIMA RMSE of 0.355 beat the naive forecast's 0.550 - about 35% better - and all five errors fell inside the 95% band, so coverage held. That out-of-sample win, not the in-sample fit, is the real evidence.

3

Always ask: price or return?

On price levels an ARIMA forecast just echoes the last value - the naive forecast - so any apparent skill is illusory. The honest evaluation is out-of-sample RMSE on returns, benchmarked against naive.

Week 4 complete — the full ARIMA and SARIMA pipeline, selected by information criteria and proven out of sample.